THE GLOBAL CATASTROPHIC RISK INDEX

Navigating Risk in an Uncertain Future

2025/2026



Augusto Lopez-Claros & John Miller
with contributions from Aminah Malik & Ben Sweeney

Table of Contents

Executive Summary 3
Introduction 5
Methodology 8
Pillar I: Quality of Governance 12
Pillar II: Health & Human Security 16
Pillar III: Economic Stability 20
Pillar IV: Gender Inequality 24
Pillar V: Environmental Vulnerabilities 28
Pillar VI: Education and Skills 32
Pillar VII: Business Environment Resilience 36
Pillar VIII: Exogenous Vulnerabilities 41
Overview of Findings 46
Areas to watch 52
Conclusion 54
Appendices 55
Bibliography 64
About the Authors 67

Acknowledgment

The authors would like to thank Aminah Malik and Ben Sweeney for their research, data collection, and drafting contributions to this report. Additional support was provided by Raghav Akula and Yasmina Mata. We would also like to thank Rumtin Sepasspour and Global Shield for their valuable input and feedback during the drafting process. The contributions of these colleagues are warmly appreciated, and this report is better as a result of their wholehearted support.

Executive Summary

Global catastrophic risk is an unavoidably complex topic. It encompasses an array of distinct risks — from environmental disaster to political strife, social unrest, and economic collapse — as well as an intricate interplay between sub-national, national, and international systems. Risks are often linked, and crises often compound one another, with vulnerabilities such as health disparities, human security issues, and competing visions of governance creating a complex web of risks.

This second issue of the Global Governance Forum's Global Catastrophic Risk Index (GCRI) seeks to distill this complexity and present a clear picture of risk in the world today. To do this, the Index assesses 163 countries across 109 indicators using data from The World Bank, the IMF, the United Nations, and other reputable sources. The findings are optimistic in their highlighting of states' capacity for mitigation and resilience but are sobering in laying bare the implications of complacence or inaction. The Index is retrospective in its analysis of past trends and historical case studies, is present in its snapshot of the contemporary catastrophic risk landscape and is forward-looking in its prescriptions for a safer and more resilient future.

The Index demonstrates not only that no country is free of risk but also that policymakers globally have often failed to take collective action against systemic and environmental risks. There is a clear correlation between countries' general level of development and their vulnerability to catastrophic risk, with poor countries already weakened by ineffective or failing governments, low economic activity, and gender and class-based inequities much more at risk. In illustration, the 2025/2026 GCRI finds the most at-risk countries in the world to be Yemen, Afghanistan, Haiti, South Sudan, and Sudan. In contrast, the least-at-risk countries are Norway, Denmark, Sweden, Switzerland, and Iceland.

> **...policymakers globally have often failed to take collective action against systemic and environmental risks.**

The impact of risks identified in the Index can be multiplied both by governments' failure to recognize them quickly, let alone preemptively, and their inability to respond effectively. In the highly integrated modern world, significant systemic shocks are unlikely to be contained within their country of origin. Instead, they will propagate among regional neighbors and trading partners with the potential to create widespread, systemic crises. It is thus critical for policymakers and citizens alike to understand the interactive and reinforcing nature of the risks facing their countries and regions to avoid the governance and societal failures that could prove disastrous for the global population.

The 2025/2026 GCRI makes it clear: in a world characterized by unprecedented uncertainty and complexity, the ability to anticipate and adapt to a multitude of shifting and emerging threats will determine which states emerge as resilient and which are left struggling to respond. It is our belief that only through a robust understanding of risk can one confront it – whether from the standpoint of a policy, civil society, academia, or individual behavior. As we continue to refine our understanding of global catastrophic risks, this report aims to be a guiding resource for policymakers, businesses, and individuals seeking to navigate the turbulent years ahead.

Introduction

The two and a half years since the publication of the Global Governance Forum's inaugural Global Catastrophic Risk Index have highlighted the rapidly evolving landscape of global catastrophic risk. Russia's invasion of Ukraine, war between Israel and Iran and its proxies, extreme weather events around the globe, the continued retreat of democracy, and the aftershocks of the COVID-19 pandemic have left many in the global community feeling a growing sense of precarity.

Daily news reports continue to illuminate the compounding and interconnected nature of risks that defy national borders and hamstring even the most capable governments. It is thus critical for authorities to recognize that risks today come in complex forms, interacting and reinforcing each other, and carry the possibility of major governance and societal failures resulting in severe consequences for the global population. The world has become undeniably interconnected, creating a possibility of reaping collective benefits from globalized commerce and exchange, but also an impossibility of insulation from instability.

To that end, the 2025/2026 Global Catastrophic Risk Index builds on the success of its 2022 predecessor and presents a report that is richer in its content and structure, in addition to being more comprehensive in its country and indicator coverage. The report remains committed to providing a comprehensive assessment of each country's relative positioning with respect to common threats.

By examining a wide array of quantitative and qualitative factors — including economic, political, social, commercial, environmental, and exogenous indicators — the GCRI seeks to shed light on the countries where vulnerabilities are most acute and to point toward those countries which have best positioned themselves for stability and resilience. At its core, the Index functions as a decision-making tool aimed at helping governments, businesses, and civil society understand the myriad risks that lie ahead and prioritize strategies for mitigation and adaptation.

This Index maintains the fundamental emphasis of its predecessor: that risks cannot be considered distinct and must be understood as interconnected, interdependent, and compounding. The economic implications of climate-induced disruptions, for instance, are now routinely entangled with political unrest and migration crises. Meanwhile, the erosion of governance capacity in fragile states continues to serve as both a symptom and a driver of broader regional instability, making it increasingly challenging to disentangle the root causes of risk.

The 2025/2026 GCRI covers 163 countries — up from 118 in 2022 — accounting for nearly 99% of global GDP and 98% of the global population. Each country is scored across eight primary risk categories — governance, health, economics, gender equality, environment, education, business climate, and exogenous indicators — capturing both internal vulnerabilities and resilience factors. By leveraging 109 unique indicators from multiple trusted sources — up from 85 in 2022 — including the World Bank, the United Nations, and he International Monetary Fund, the GCRI integrates nearly 20,000 data points to provide a nuanced yet accessible view of global risk.

The 2025/2026 Index improves on the 2022 edition by adding Health & Human Security as a stand-alone pillar, owing to the centrality of health to a country's overall risk profile that has been on display since the emergence of the COVID-19 virus. Technological indicators have also been added as a category of exogenous vulnerabilities, as the global proliferation of information technologies has become a critical component of how the global community interfaces and exchanges. Other changes include an array of indicator additions and substitutions (see Figure 2).

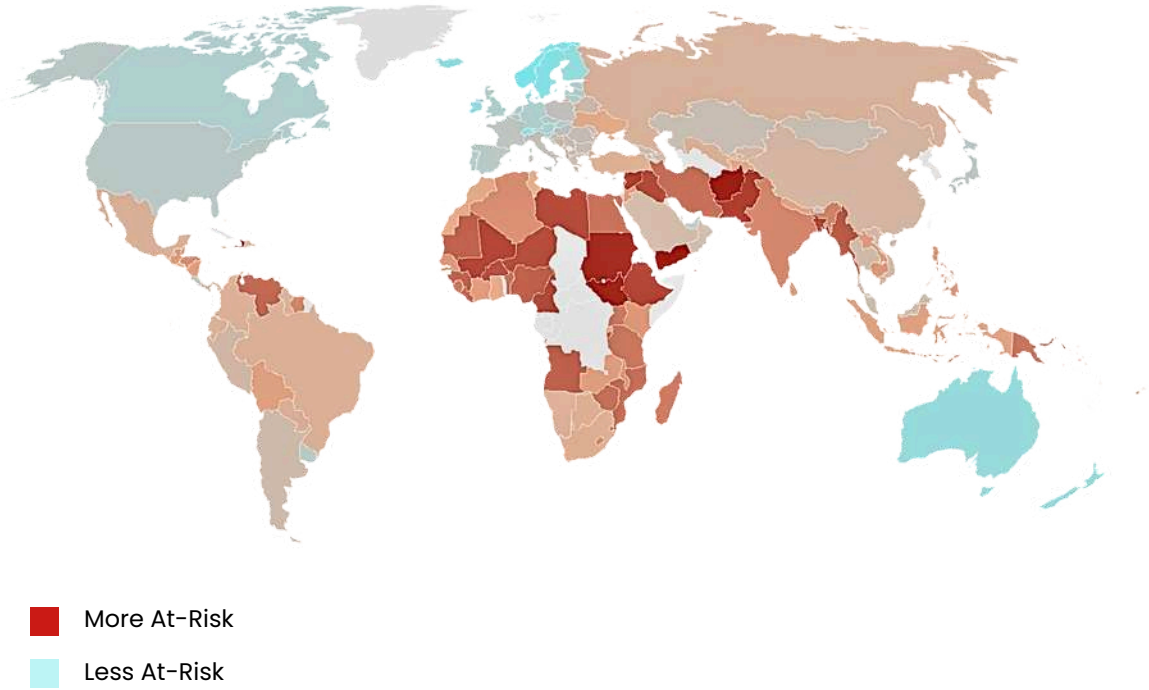
Compared to the top 10 most at-risk countries in the 2022 Index, five remain the same (Afghanistan, Mali, Pakistan, Sudan, and Yemen), while five countries have entered the top 10 (Democratic Republic of Congo, Ethiopia, Haiti, South Sudan, and Syria). Among the top 10 least at-risk countries, seven remain the same

(Australia, Austria, Denmark, Finland, Ireland, Norway, and Sweden), while three countries are new to the lowest-risk group (Iceland, New Zealand, and Switzerland).

The scores for each indicator highlight the degree to which countries are vulnerable — whether through poor policy, bad geography, or bad luck or a combination thereof — to risks that could result in catastrophe. Addressing vulnerability to these risks requires action on two levels. Some risks can be managed internally through national strategies and reforms, where indices like this one can help guide domestic policy and initiatives. Other risks, however, are beyond the capacity of any single country to mitigate and demand a coordinated regional or global response. In such scenarios, individual nations may only be able to focus on minimizing their exposure and bolstering resilience against these external threats.

The 2025/2026 GCRI makes it clear: in a world characterized by unprecedented uncertainty and complexity, the ability to anticipate and adapt to a multitude of shifting and emerging threats will determine which states emerge as resilient and which are left struggling to respond. As we continue to refine our understanding of global catastrophic risks, this report aims to be a guiding resource for policymakers, businesses, and individuals seeking to navigate the turbulent years ahead.

Figure 1. The Geography of Global Catastrophic Risk



Countries not included in the 2025/2026 GCRI are shaded in grey.

Methodology

In this edition of the Global Catastrophic Risk Index, an attempt has been made to be comprehensive in coverage, subject to data limitations and the requirements that, for a country to be included, sufficient data be available across the 109 indicators that are incorporated into the Index and, for an indicator to be included, data be available for a broad enough set of countries. This Index aggregates nearly 20,000 individual data points into a set of eight pillars, weighted equally for simplicity. Despite the scale and intention to be comprehensive, data-related challenges unavoidably affect the granularity and precision of indices like this one. However, the primary utility of the GCRI lies in its ability to provide a multifaceted, structured, comparative framework for assessing risk and resilience across a broad sample of risk factors and countries. However, it is important to acknowledge limitations in the data that resulted in limited country coverage and the exclusion of certain countries likely to be at high risk.

The current reality in terms of global datasets that try to capture various risk factors is that there is a wealth of data on macroeconomic and economic development indicators, reflecting the important investments in data collection made in this area by international financial institutions. However, there is considerably less data on environmental and social factors, not to mention other vulnerabilities that are inherently more difficult to measure, particularly in the context of middle and low-income countries. In some cases, a lack of data has resulted in a reliance on proxies which the authors believe to be suitably accurate and descriptive of the item under measurement, or in the exclusion of the country altogether. In this Index, countries are only included if data is available for at least 82 (or 75%) of the 109 indicators used in the Index and a score is able to be calculated for each pillar.

Among those countries not included in the Index due to data unavailability were a number of small island states including Antigua and Barbuda, Dominica, Kiribati, Micronesia, Nauru, Palau, St. Kitts and Nevis, and Tuvalu. In addition to these island states, small countries (in land area and/or population) such as San Marino, Andorra, and Palestine were excluded due to data availability issues. In many cases, these countries suffer from limited resources and institutional capacity for comprehensive data collection. Compounded by their small populations and geographical isolation, these factors make it costly and logistically challenging to maintain robust statistical systems. Even when data is collected by international organizations, these nations are often overlooked in global data initiatives due to their minimal impact on large-scale economic or geopolitical metrics.

Anecdotally, it is evident that even small environmental, economic, or governance shocks can have disproportionately devastating impacts on these countries. With their limited populations, economies, and land areas, there is virtually no margin for error, making precarity the norm rather than the exception. Unfortunately, the lack of reliable and comprehensive data prevents their inclusion in the 2025-26 GCRI. In future editions of the Index, the authors are committed to exploring methods to better capture and represent the risks faced by these small but highly vulnerable nations.

Figure 2. Structure of the Global Catastrophic Risk Index

Pillar I | Quality of Governance

1. Building Blocks of Good Governance
   a. Political Stability & Absence of Violence
   b. Control of Corruption – Perceptions
   c. Voice & Accountability
   d. Rule of Law
   e. Government Effectiveness
   f. Access to State Jobs by Social Group
2. Regime Fragility
   a. Factionalization of Elites
   b. Group Grievances
   c. Regime Type
   d. Presence of Conflict
   e. Political Terror
3. Incidence of Corruption
   a. Corruption Perceptions Index
   b. Reliability of Police Services
   c. Judicial Independence
   d. Electoral Participation
4. Information Freedoms
   a. Political Rights
   b. Press Freedom
   c. Freedom of Thought, Conscience, & Religion

Pillar II | Health & Human Security

1. Healthcare Access & Quality
   a. Basic Hygiene
   b. Doctors per Capita
   c. Reproductive Health
   d. Average Cost of Basic Medical Care
   e. Service Coverage
   f. Routine Vaccinations
2. Population Health & Risk Factors
   a. Obesity Rate
   b. Child Malnutrition Rate
   c. Avoidable Mortality
   d. Alcohol Consumption
   e. Dietary Risks
   f. Smoking Prevalence
3. Health Status Indicators
   a. Life Expectancy at Birth
   b. Under-5 Mortality Rate
   c. Infant & Maternal Mortality Rate
   d. Chronic Diseases
   e. Infectious Diseases
4. Security
   a. Intentional Homicides
   b. Human Trafficking
   c. Sexual Assault[1]
   d. Serious Assault

Pillar III | Economic Stability

1. Macroeconomic Indicators
   a. Budget Deficit
   b. Debt Level
   c. Inflation Rate
   d. Labor Force Participation Rate
   e. Unemployment Rate
   f. Total Debt Service
2. Structural Resilience
   a. Size of the Underground Economy
   b. Income Inequality
   c. Natural Resource Endowment
   d. Quality of Infrastructure
   e. Integration with Global Markets
   f. Trade Restrictiveness
3. Financial Resilience
   a. Foreign Currency-Denominated Loans to Total Loans
   b. Liquid Assets to Short-Term Liabilities
   c. Non-Performing Loans to Total Gross Loans
   d. Regulatory Capital to Risk-Weighted Assets

Pillar IV | Gender Inequality

1. Employment and Autonomy
   a. Entrepreneurship Access
   b. Incentives to Work
   c. Property Rights
   d. Working Restrictions
   e. Labor Force Participation Rate - Gender Disparity
2. Embedded Obstacles
   a. Legal Restrictions
   b. Mobility Restrictions
   c. Legislation on Domestic Violence
   d. Legal Restrictions on Marriage and Divorce
   e. Proportion of Seats Held by Women in National Parliaments
3. Norms and Decision-Making
   a. Power Relationships at Home
   b. Violence Against Women

---

1. Sexual assault is included here rather than under the gender equality pillar as that pillar is intended to capture gender disparity, embedded obstacles, and restrictions. Violence against women (included in that pillar) is women-specific, while we treat sexual assault as a security issue that is not exclusively limited to women victims and tracks closely with homicides, trafficking, and other serious assaults which are analyzed here.

Pillar V | Environmental Vulnerabilities

1. Environmental Policy
    a. Pesticide Use
    b. Air Pollution
    c. Renewable Energy Sourcing
    d. Fossil Fuel Subsidies
    e. Overall Post-Tax Subsidies
2. Geography
    a. Agricultural Land
    b. Land Area Below 5m
    c. Urban Population
3. Disasters
    a. Volcano-Related Injuries and Deaths
    b. Temperature Change
    c. Tsunami-Related Injuries and Deaths
    d. Droughts, floods, extreme temperature events
    e. Internally Displaced by Natural Catastrophes

Pillar VI | Education and Skills

1. Quantity of Education
    a. Primary Enrollment
    b. Secondary Enrollment
    c. Tertiary Enrollment
    d. Mean Years of Schooling (Primary and above)
2. Quality of Education
    a. Educational Attainment (Primary)
    b. Educational Attainment (Upper Secondary)
    c. Educational Attainment (Bachelor)
    d. Adult Literacy Rate
3. Job Training
    a. Adult Education and Job Training
    b. Brain Drain

Pillar VII | Business Environment Resilience

a. Property Rights Regime
b. Facilitating Entrepreneurship
c. Getting credit
d. Paying Taxes
e. Enforceability of Contracts

Pillar VIII | Exogenous Vulnerabilities

1. Demographics
    a. Population Density
    b. Population Growth Rate
    c. Coastal Population
2. Technology
    a. Access to Internet
    b. Cell Phone Usage
    c. Cellular Coverage
3. Defense
    a. Security Threats
    b. External Influence
    c. Nuclear Capacity
    d. Military Spending
    e. Deadliness of Conflict
    f. Danger of Conflict to Civilians
    g. Geographic Diffusion of Conflict
    h. Armed Group Fragmentation



PILLAR I

Quality of Governance

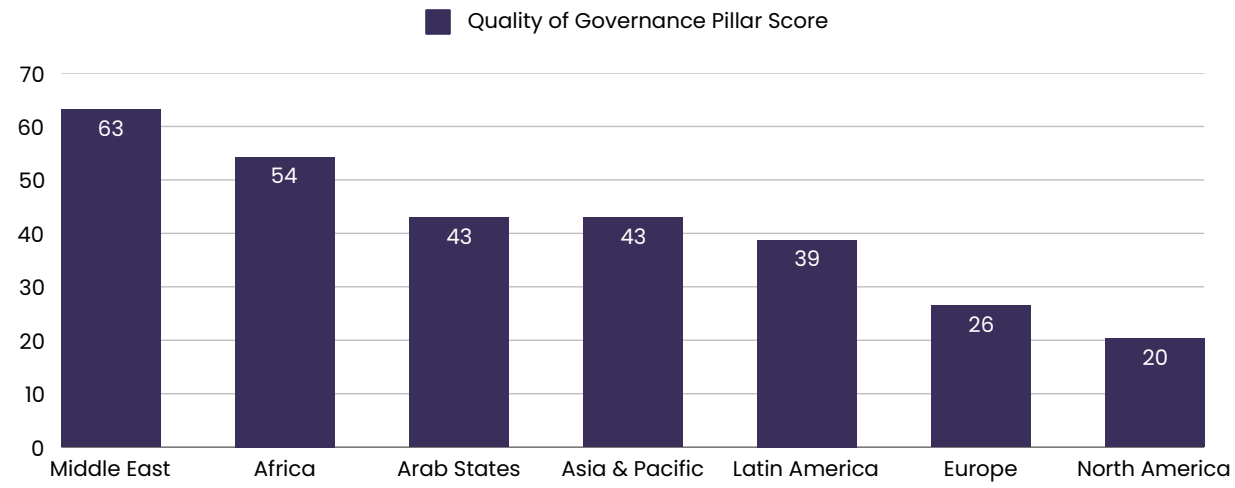
Quality governance is a core pillar determining countries' stability, resilience, and potential for widespread prosperity. A stable, effective government forms the backbone of a country's ability to mitigate crises and leverage opportunities. Quality of governance is therefore not only a crucial measure of risk in itself but also an important indicator of a state's capacity to address risks explored elsewhere in the Index. Countries with robust governance structures are better equipped to navigate challenges, maintain stability, and foster long-term growth.

Effective governance ensures the proper functioning of institutions and thus directly influences a country's economic resilience, social cohesion, and developmental trajectory. Effective governments transparently and efficiently steward resources and provide services to their citizenry, while officials are held accountable through political and legal means. By contrast, weak governance heightens risk by engendering uncertainty, in turn deterring investment, stifling growth, and increasing a country's vulnerability to external shocks.

The global governance landscape is ever-changing and has undergone significant changes in the past several years. While some regions have made strides toward improving governance, others face growing risks and challenges. Countries in Europe and North America continue to rank highly on governance indicators such as rule of law, political stability, and government effectiveness. Moreover, many East Asian countries have strengthened their institutions, contributing to rapid economic growth and development. However, parts of Latin America, Africa, and the Middle East face persistent risk as corruption, fragile regimes, conflict, and political instability undermine development progress and weaken social cohesion.

Globally, concerns over rising authoritarianism, erosion of democratic institutions, and increasing restrictions on political rights are well-founded.

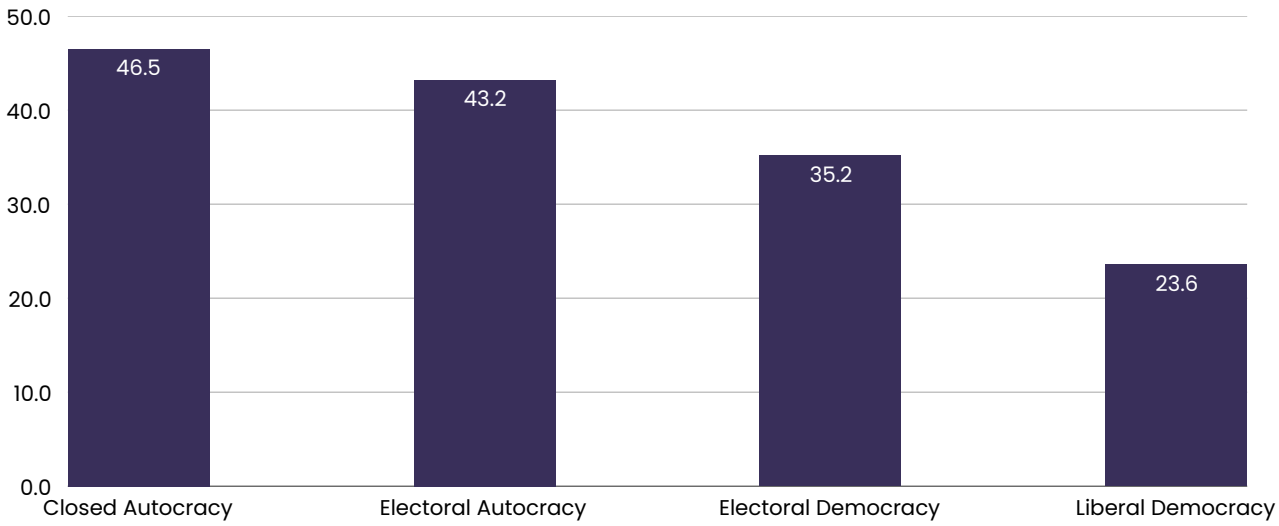
Figure 3: Governance Risk by Region



Higher numbers represent higher risk.

Globally, concerns over rising authoritarianism, erosion of democratic institutions, and increasing restrictions on political rights are well-founded. In 2023, The Economist found that less than 8% of the world's population live in a fully democratic country, while nearly 40% live under authoritarian rule.[2] This trend is only worsening as authoritarianism is on the rise, particularly in parts of Africa, Asia, and Eastern Europe.

Figure 4: GCRI Score by Regime Type



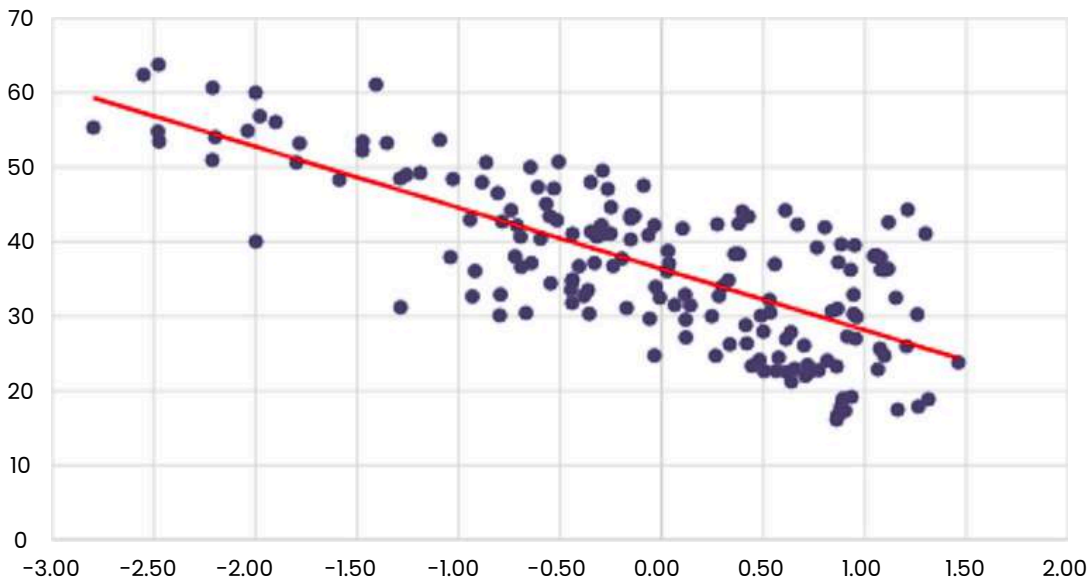
The foundational aspects of effective governance are political stability and the absence of violence. In the absence of either of these, first-order political crises may arise wherein existential risk is magnified such that other indicators lose relevance. To illustrate, a state consumed by regime change or sectarian violence can hardly afford to focus its attention on improving income distribution, increasing secondary school attendance, or focus on sourcing its energy from renewable sources.

Stability is intimately related to the rule of law, which is characterized by a system in which all individuals, businesses, and government officials are subject to the same legal standards. Reliable policing, judicial independence, control of corruption, and electoral participation all point to the rule of law. A strong legal framework that enforces property rights, anti-corruption measures, and accountability for unlawful actions fosters a business-friendly environment, reducing risk and leading to economic growth. Conversely, countries with weak rule of law struggle to attract investment, as legal uncertainty deters business activity. In countries where the rule of law is compromised, such as in parts of sub-Saharan Africa, the private sector is often underdeveloped, and economic growth remains sluggish.

Corruption is one of the most common and persistent ills plaguing countries around the world. Corruption heightens risk by undermining the rule of law, yielding inefficiencies, distorting incentives, and discouraging private sector innovation and development. It also subverts government revenue capture and, as a result, limits the ability of the government to invest in productivity-enhancing areas, such as education, infrastructure, and health. In countries like South Sudan, where corruption is a systemic issue that undermines development, it leads to severe poverty, inequality, and instability. On the other hand, nations that have made significant efforts to reduce corruption, such as Botswana, have experienced more stable and sustained development.

2. The Economist. "Democracy Index 2023".

Figure 5: GCRI Score x Political Stability & Absence of Violence



Source: The World Bank

Effectively responding to the risks and obstacles facing one's country first requires understanding what those risks and obstacles are. Transparency leads to accountability. In the political sphere, enhanced accountability helps ensure that governments are responsive to citizens' needs and less prone to corruption. The better individuals understand the realities of their country and its position in the world, the better they can orient themselves to remedy ills and take advantage of opportunities. A media free from intimidation and state restrictions is critical to fostering this knowledge. Without a free press, however, authoritarian regimes can manipulate information, hide corruption, and stifle dissent, leading to economic mismanagement and instability.

Effective governance is essential for countries' stability, prosperity, and development. Countries that prioritize political stability, champion the rule of law, ensure government effectiveness, and promote information freedoms are more likely to experience sustained economic growth and resilience to shocks. Conversely, countries plagued by corruption, political terror, regime fragility, and restricted freedoms face greater economic, political, and social vulnerabilities.

CASE STUDY

POLITICAL STABILITY IN BARBADOS & HAITI

Political stability — marked by durable and legitimate institutions, the absence of violence, peaceful transition of power, and the rule of law — plays a critical role in mitigating the risks a country and its citizens face. A stable political foundation enables a state to make decisions in the best interests of its citizens over the short, medium, and long term. It creates a confident business environment, enables the provision of public services, and can effectively mobilize resources to head off crises. The importance of such stability can be seen in two contrasting examples among Caribbean neighbors —Barbados and Haiti.

The present crisis in Haiti is made worse by the incapacity of its government. The country's struggle with failed governance has left its citizens fending for themselves as the state is unable to perform the most basic of functions. The assassination of President Jovenel Moïse in 2021 proved to be the blow that crippled an already fragile government, creating a power and service vacuum that has fueled the rise of armed gangs that now control large parts of the country's capital. In addition to the security threats presented by gang violence, the lack of central authority has led to lawlessness more broadly, food insecurity, limited healthcare, and a fractured economy rife with corruption. Under gang rule, the statistics of state failure are staggering: over 5.5 million citizens need humanitarian assistance, 1.4 million are on the brink of famine, 700,000 have been displaced, and 3,500 have been killed since the beginning of the year.

The government's inability to manage public resources, protect its citizens, or deliver essential services has resulted in an impoverished population largely left to fend for itself, with communities increasingly reliant on local gangs or international aid for basic needs. Basic state functions such as health care, education, law enforcement, and judicial systems have either collapsed or been co-opted by local armed groups, further weakening the rule of law and fueling corruption; since 1980, GDP per capita (in constant dollars) has declined by more than 43%. Until a legitimate government takes power, establishes the rule of law, and gets the economic wheel turning, stagnation and decline will be the rule.

On the other hand, Barbados showcases the benefits of political stability in managing risks. The country of 281,000 has long benefitted from high levels of public trust in its institutions. Since gaining its independence in 1966, the country's strong rule of law, tradition of coalition-building, transparent policy, and strong control of corruption have created a resilient, durable political climate. The country thus has a strong capacity to manage both internal and external risks, significantly lowering its vulnerability to catastrophic risk. As one of the most politically stable Caribbean nations, Barbados has avoided the social unrest and political volatility that affect many of its regional neighbors. The country holds regular, peaceful elections, and its government is underpinned by a robust legal and constitutional framework that ensures the separation of powers and protects civil liberties. The country has also invested in education, healthcare, and social welfare, which has contributed to a relatively high standard of living. These factors, combined with careful economic management and a focus on tourism and financial services, have enabled Barbados to grow and prosper.

PILLAR II

Health & Human Security



The COVID-19 pandemic profoundly altered global perceptions of health and human security. It exposed the vulnerabilities of even the most developed healthcare systems and illuminated health's impact on a range of economic, social, and political risks. To illustrate, the pandemic killed nearly 15 million people globally and precipitated a global economic contraction of 3.3% in 2020 (the worst global recession since World War II).[3,4]

It is therefore clear that Health & Human Security must stand alone as its own pillar in the GCRI. In this year's report, we have adapted and expanded the set of indicators included as Exogenous Vulnerabilities in 2022 to more accurately capture the broad set of indicators that comprise a country's health risk profile. These include measures of healthcare affordability, accessibility, and quality; population health and risk factors; nationwide health status indicators; and measures of human security including homicide, trafficking, and assault rates.

Health forms the bedrock of citizens' capacity to engage meaningfully and productively in the world around them.

Health forms the bedrock of citizens' capacity to engage meaningfully and productively in the world around them — including in national economies, local communities, and in their families. Health is an enabler of self-actualization, longevity, and productivity.

The establishment of fit-for-purpose healthcare infrastructure is fundamental to mitigating a range of risks. In recent decades, health infrastructure around the world has been augmented to ensure broader coverage, a greater array of services, and increased funding for research. For instance, the under-5 mortality rate – an indicator used in the GCRI as a proxy for healthcare coverage and quality – has declined 60% since 1990, from 93 deaths per 1,000 live births to 37.

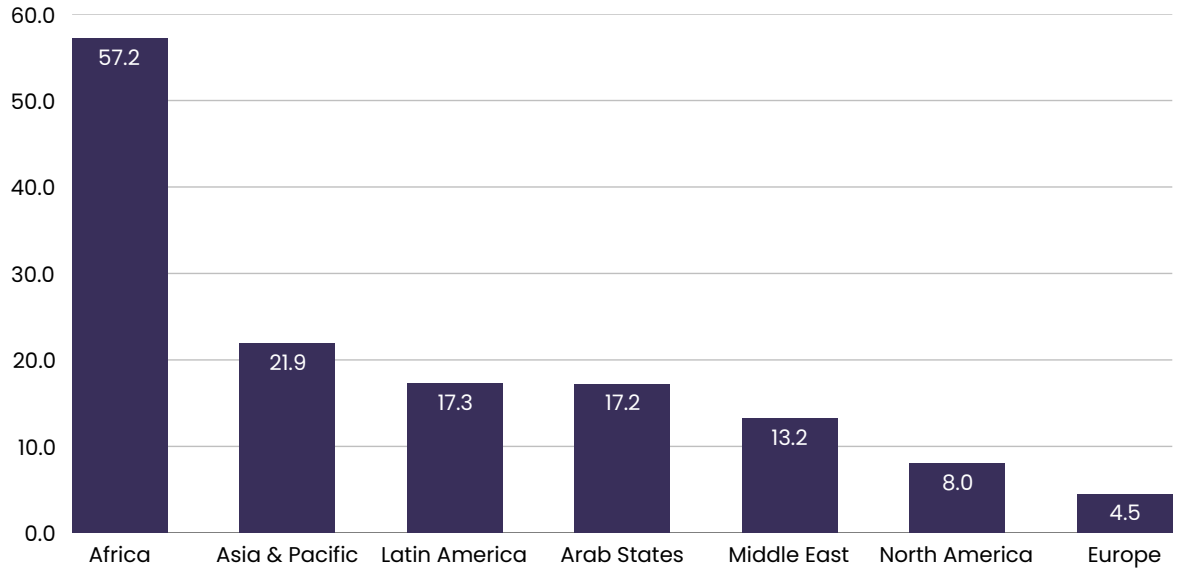
However, as the global response to the COVID-19 pandemic demonstrated, there is still vast progress that must be made, particularly in developing countries. According to United Nations estimates, over 18 million additional health workers will be needed by 2030 to ensure high-quality, accessible coverage.[5] The risk and resulting need is greatest in those areas that are currently underserved. In addition, the pandemic further underscored the inequities in medical coverage that exist across countries, regions, and socio-economic groups. To return to the under-5 mortality rate example, the top 20 countries with the highest rates were in Africa, where the continent's average under-5 mortality rate is nearly twice the global average at 73 deaths per 1,000 live births.

3. Center on Budget and Policy Priorities. "Tracking the Recovery from the Pandemic Recession." CBPP, www.cbpp.org/research/economy/tracking-the-recovery-from-the-pandemic-recession.

4. Johns Hopkins University. "COVID-19 Dashboard by the Center for Systems Science and Engineering (CSSE)." Coronavirus Resource Center, coronavirus.jhu.edu/map.html.

5. World Health Organization. Standards for Improving Quality of Maternal and Newborn Care in Health Facilities. World Health Organization, 2016, apps.who.int/iris/bitstream/handle/10665/250368/9789241511131-eng.pdf?sequence=1.

Figure 6: Under-5 Mortality Rate



Source: The World Bank

The pandemic led to widespread disruption in healthcare delivery, delayed routine health services, and worsened health outcomes in many countries around the world. Routine vaccination programs for diseases like measles and polio were halted in numerous low-income countries, creating a resurgence of preventable illnesses. In developed nations, healthcare systems have struggled with burnout among health professionals and capacity constraints, leading to a backlog of untreated patients.

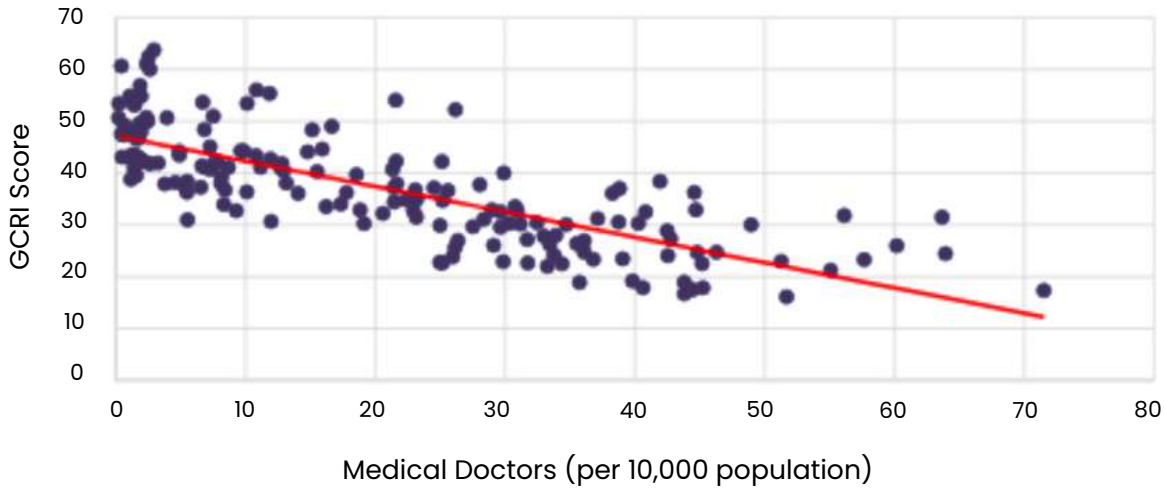
On a more positive note, the pandemic has also accelerated advancements in telemedicine and digital health, enabling remote consultations and expanding access to healthcare. Moreover, the rapid development and distribution of COVID-19 vaccines demonstrated the potential for global cooperation in addressing health crises.

Basic hygiene is a prerequisite for health. Countries that lack universally accessible handwashing facilities are more vulnerable to the spread of infectious diseases, significantly increasing the risk of outbreaks that can destabilize communities and overwhelm healthcare systems. For example, sub-Saharan Africa and parts of South Asia continue to struggle with access to basic sanitation. This not only leads to higher rates of preventable disease but also hampers development by increasing avoidable healthcare costs and reducing workforce productivity.

Beyond hygiene, vaccinations are also key to preventing the spread of infectious diseases, reducing mortality rates, and minimizing strain on healthcare systems. Countries with high vaccination rates are less vulnerable to epidemics and more capable of maintaining public health stability. For example, countries with established immunization programs, such as Japan and Australia, have been able to control outbreaks of vaccine-preventable diseases even during the pandemic. In contrast, regions with low vaccination rates, such as parts of Africa and Southeast Asia, have experienced resurgences of diseases like measles and polio.

The number of doctors per capita is a crucial indicator of a country's healthcare capacity. Nations with higher ratios of healthcare professionals are better equipped to provide timely care, manage chronic conditions, and respond to public health emergencies. Conversely, countries with low doctor-to-patient ratios face significant challenges in providing adequate care, particularly during crises. Globally, there is an average of 17.2 medical doctors per 10,000 population. However, significant disparities exist: while 77 countries have fewer than 10 medical doctors per 10,000 population, 53 countries have more than 30 medical doctors per 10,000 population.

Figure 7: GCRI x Medical Coverage



Source: World Health Organization

Independent of the strength of a country's healthcare system are a series of population risk factors that heavily factor into a country's health outcomes. For instance, the prevalence of lifestyle-related health risks such as obesity, alcohol consumption, and smoking, can increase the prevalence of non-communicable diseases like heart disease, diabetes, and cancer. Conversely, high levels of malnutrition and food insecurity can increase susceptibility to illness, stunt development, and cripple individual productivity.

Turning away from health indicators, the category of human security provides several critical indicators of the population's well-being. Violence, trafficking, and exploitation diminish victims' welfare, undermine social cohesion, disrupt communities, and create environments of fear and instability. Countries with high rates of trafficking and assault often struggle with weak governance and inadequate law enforcement, increasing the risk Turning away from health indicators, the category of human security provides several critical indicators of the population's well-being. Violence, trafficking, and exploitation diminish victims' welfare, undermine social cohesion, disrupt communities, and create environments of fear and instability. Countries with high rates of trafficking and assault often struggle with weak governance and inadequate law enforcement, increasing the risk of organized crime and terrorism. Countries in conflict zones, such as parts of the Middle East and Central America, experience high rates of human trafficking and gender-based violence. Combating these issues requires strong legal frameworks, coordinated law enforcement efforts, and community-based interventions to protect vulnerable populations.

CASE STUDY

HEALTH WORKER DENSITY IN BOLIVIA & URUGUAY

The human element of health coverage is a critical component of national health infrastructure and the key to service delivery. As such, the density of health workers in a given population is an effective indicator for a range of healthcare outcomes. In the wake of the COVID-19 pandemic, the centrality of health workers to a country's catastrophic risk profile has been more widely understood, but major disparities persist. Two South American countries, Uruguay and Bolivia, present a stark contrast: while Uruguay boasts one of the highest doctors to population ratios in Latin America, Bolivia struggles with a much lower level.

Uruguay's 46 doctors per 10,000 citizens is among the highest in Latin America, while Bolivia's 13 doctors per 10,000 citizens is among the lowest. The impact of this disparity extends throughout citizens' lifespans. For instance:

- The under-5 mortality rate is more than 3.5 times higher in Bolivia than in Uruguay;
- Just 69% of Bolivian children receive routine vaccinations, whereas 94% of Uruguayan children receive such vaccinations;
- In Bolivia, more than 33% of all deaths due to non-communicable diseases are classified as premature. In Uruguay, the figure is roughly 25%; and
- Bolivian life expectancy is 65, whereas it is 20% higher among Uruguayans (78).

In Uruguay, a high density of healthcare professionals and modern health infrastructure contribute to lower rates of preventable hospitalizations for chronic conditions such as diabetes and heart disease. In contrast, Bolivia has one of the highest rates of deaths from chronic conditions in Latin America, with its healthcare system struggling to match patients with providers and to sufficient access to specialized care.

Uruguay's success is due in part to the establishment of a National Integrated Health System in 2007, which expanded access to healthcare by integrating public and private providers. The System also created an accessible and affordable health system for all citizens, regardless of income or social status. The sustainability of the system and attractiveness to healthcare providers is ensured by funding through a national fund that is paid into by employers, employees, and the government.

In Bolivia, the combined effects of an underfunded healthcare system, an uneven distribution of healthcare resources, and limited access to medical training have hindered service delivery and yielded a shortage of healthcare workers, particularly in rural areas. The country's United Health System, established in 2019, has run into significant budget shortfalls that have precluded the hiring of more health workers. Even where there are doctors, the country's rural areas frequently lack even basic healthcare facilities, leading to a concentration of doctors in urban centers, which in turn results in significant disparities in access to care levels across the country.



PILLAR III

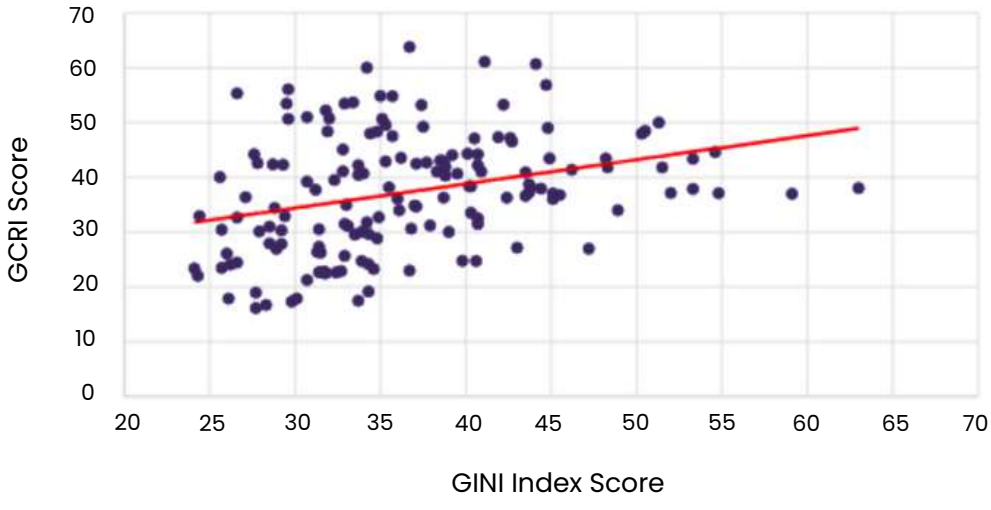
Economic Stability

Macroeconomic stability is a critical element for the successful functioning of a country. It is a precondition for the full capacity of a state to exercise its fundamental mandates of ensuring the security and well-being of its citizens, maintaining law and order, and providing necessary social services. In 2022, we noted that there are no known instances of countries that have managed to grow sustainably while pursuing imprudent fiscal policies that have, for example, fueled inflation and exchange rate instability and contributed to the emergence of various macroeconomic imbalances. This observation remains true. Prudent fiscal and monetary policy, coupled with effective stewardship of a nation's finite resources, increases business and consumer confidence. These, in turn, stimulate domestic industries, drive entrepreneurship and foreign investment, promote job creation, and enhance social welfare. Volatile inflation, pronounced income inequality, fluctuating exchange rates, large fiscal deficits, and other risks explored in the Index can erode a country's wealth, reduce its global competitiveness, and ultimately destabilize a country politically. At their worst, these spiraling externalities can undermine the basis of democracy and social order, leading to violence.

High and volatile inflation presents a critical risk by eroding savings, hindering investment, and disproportionately affecting the poor, who spend a larger share of their income on
essential goods. Countries experiencing very high inflation, like Venezuela, or highly variable inflation, like Argentina, face insurmountable challenges in stabilizing their economies. Inflation not only causes economic dislocation but also risks undermining social and political stability because it disproportionally affects lower income groups; unrest is often inevitable when wages fail to keep pace with rising prices, or it tends to lead to repression and violence in countries with authoritarian forms of governance

Another key indicator of a country's economic health is its unemployment rate. A low unemployment rate indicates low levels of idle capacity and high levels of economic engagement. Conversely, a high unemployment rate signals underutilized productive capacity, leading to diminished economic growth. Widespread unemployment also contributes to social and political tension. When individuals who wish to work are unable to secure gainful employment, discontent mounts and may result in increased crime, protests, and political instability. For example, 28% unemployment in South Africa has resulted in large scale workers´ protests across the country. Similarly, high levels of income inequality can be destabilizing. Stark disparities in economic attainment foster social discontent, tension, and resentment that can lead to unrest. Conversely, countries with lower levels of income inequality often benefit from a greater degree of social cohesion and stability.

Figure 8: GCRI Score x Income Inequality



Source: The World Bank

As the recovery from the global financial crisis showed a decade ago and the recovery from the COVID-19 pandemic demonstrated more recently, sound public finances and fiscal sustainability pay dividends in difficult times. To illustrate, we can look at the example of high public debt. In 2020, public debt levels rose by roughly 28 percentage points of GDP across the world in response to the COVID-19 pandemic.[6] While some of the spending was fully justified as a response to the public health crisis, the fact is that in 2025 we see a global public finance landscape that is characterized by a much narrower fiscal space, with countries across the planet operating under much higher levels of public indebtedness, sharply limiting their ability in the future not just to respond to ongoing needs to spend in education, to finance the transition to a renewable energy economy, to invest in carbon resilient infrastructures, but also undermining their ability to deal with the next crisis. Furthermore, unlike previous decades, fiscal sustainability questions no longer apply just to countries in the developing world but have also become a feature of high-income countries and this is worrying because large countries have the potential to destabilize the whole global financial system and economy, as happened during the 2008/09 global financial crisis.

More generally, countries that have managed to sustain economic dynamism (evidenced by low debt, manageable inflation, and high labor force participation rates) have been able to allocate more resources to productivity and innovation, capture higher levels of tax revenue, and secure higher levels of investment.

A high-performing, dynamic, and well-structured economy can also help mitigate its shortcomings.

A high-performing, dynamic, and well-structured economy can also help mitigate its shortcomings. For example, Japan has a high debt-to-GDP ratio, but its advanced economy, stable banking system, and significant domestic savings help mitigate its exposure to shocks. On the other hand, many developing economies experience heightened vulnerability to debt crises owing to limited access to credit and an inability to absorb shocks such as commodity price swings.

Looking toward integration with global markets, a more outward-looking orientation has become an essential element of successful economic policy and risk mitigation as the world becomes increasingly interdependent. Greater integration with the world economy also serves as an important channel for absorbing technological advances from abroad, including improvements in management practice and positive effects on the build-up of human capital that derives from being able to tap into global systems of knowledge.

6. IMF Global Debt Database.

CASE STUDY

INFLATION IN CHILE AND VENEZUELA

High and variable inflation significantly increases a country's vulnerability to social, political, and economic crises, placing it at a higher level of catastrophic risk. By decreasing the value of the currency, high inflation disrupts financial systems and devalues savings, hence decreasing government tax revenues. This makes it harder for governments to fund public services or repay debt, leading to vicious cycles where a collapsing tax base leads to money printing which boosts inflation further and, unchecked, can lead to heightened political instability as well. The impact of inflation can be seen in the diverging stories of two South American countries — Venezuela and Chile.

In Venezuela, a decade of hyperinflation (reaching nearly 1,700,000% in 2018) has yielded an almost unparalleled crisis of unrest, emigration, and political instability. Although Venezuela is blessed with the world's largest oil reserves, it has fallen prey to the "resource curse" – an overreliance on a natural resource that has left other sectors underdeveloped. Declining oil prices, rampant corruption, and economic mismanagement have ruined what was once one of Latin America's most prosperous economies and have plunged millions into poverty.[7] Basic goods (such as food and medicine) are unaffordable or unavailable entirely, leading to widespread malnutrition, a public health crisis, and mass emigration. Democratic institutions have eroded as the government of President Nicolás Maduro tightens its grip on power in response to growing dissent, further undermining the rule of law. The inflationary crisis has transformed Venezuela into one of the most unstable and impoverished countries in Latin America.

In contrast, Chile has a track record of maintaining a relatively low and stable inflation rates (averaging 4.4% annually over the past decade). Chile has benefitted from an independent central bank that is able to control inflation through monetary policy. In addition to largely credible interest rate policy, Chile's fiscal discipline—characterized by conservative government spending, low public debt, and the use of stabilization funds from copper revenues—has helped insulate the economy from commodity price swings and other external shocks. Chile's economic stability has attracted significant foreign investment, boosted consumer and business confidence, and fostered steady economic growth. Adequate government tax revenues have allowed for more effective poverty reduction programs and freed resources for spending on infrastructure, education, and health. In short, economic stability has helped reduce poverty and improve social welfare, with Chile emerging as one of the most prosperous and stable countries in the region. Reflecting broadly sound economic policies implemented over the last several decades Chile joined the OECD, the high-income club in 2010.

7. Council on Foreign Relations. Venezuela: The Rise and Fall of a Petrostate. 26 Sept. 2023, www.cfr.org/backgrounder/venezuela-crisis.

PILLAR IV

Gender Inequality



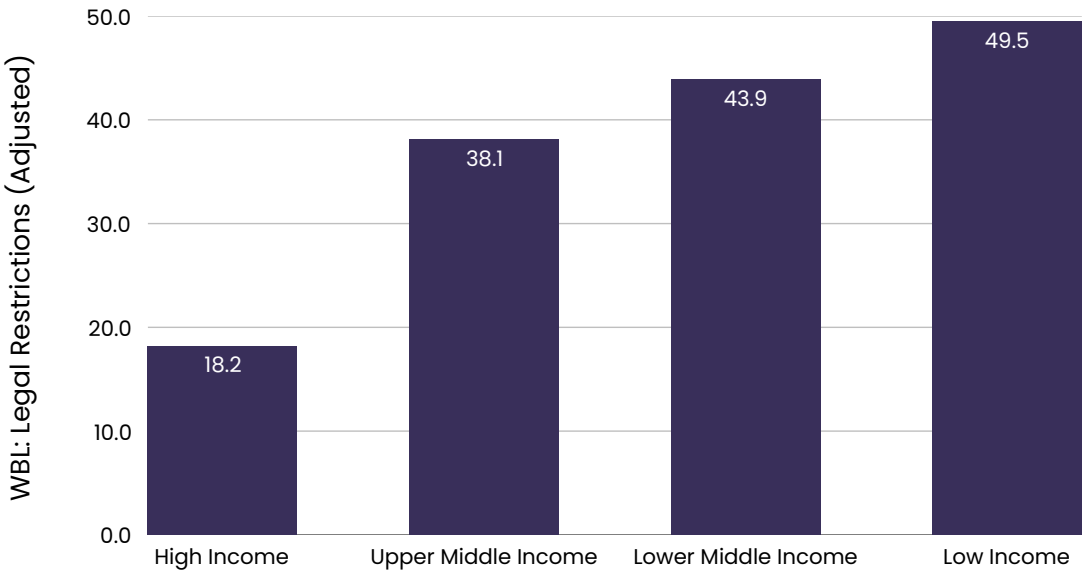
As The Global Governance Forum's Gender Equality and Governance Index makes very clear, the integration of women into political, economic, and social institutions is not just an issue of fairness; it is a core driver of a country's resilience to a range of potentially catastrophic risks. The benefits of greater gender equality include improved government effectiveness, greater rule of law, higher levels of political stability, less violence, and enhanced competitiveness.

The exclusion of women from the economy and the corridors of power only serves to heighten social tension and limit the capacity of nations to innovate and respond to crises.

In this Index, we observe that countries succeeding in integrating women into political bodies and the economy score more highly across a range of risk indicators; they exhibit stronger economic growth, greater political stability, and a more robust social fabric. Moreover, governance inclusive of women leads to better decision-making, greater transparency, and enhanced accountability. The exclusion of women from the economy and the corridors of power only serves to heighten social tension and limit the capacity of nations to innovate and respond to crises.

Efficiency in the knowledge-based international economy is dependent on a country's ability to tap into its reservoir of talents and skills, which must include women. When, because of tradition, religion, social taboos, or outright prejudice, women are prevented from contributing to the life of a nation, that nation suffers. A society that does not allow women equal access to education, property rights, and credit stunts its growth potential by halving the talent pool from which the private and public sectors can draw.

Figure 9: Legal Restrictions Against Women by Country Income Level



Source: The World Bank

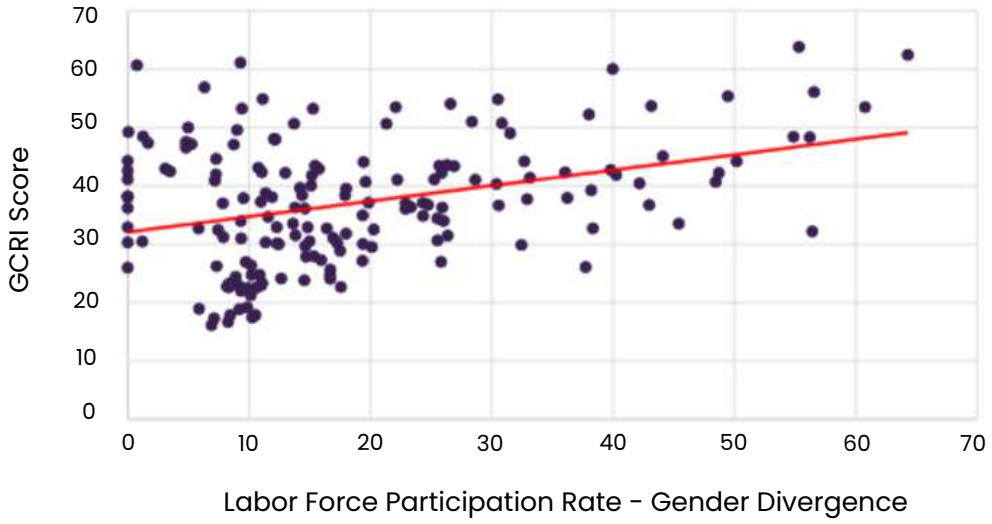
It is no surprise that the most competitive countries in the world economy, those that have been better able to operate on the boundaries of the technology frontier, are also those in which women have been given the greatest opportunities to be equal partners with men. Industry aside, without full access to property rights or credit, women cannot fully participate in economic activities such as entrepreneurship

or investment. In an era where innovation drives growth, gender-based exclusion means that countries fail to benefit from the full potential of their workforce, thus limiting their ability to compete on the global stage. Based on current labor force participation rates, achieving gender parity would add an additional 675 million women to the global workforce.

Given the clear benefits of ensuring women's full participation in the global economy, it is shocking that 178 countries continue to maintain legal barriers that prevent women's full economic participation. These include 86 countries in which women face some form of job restriction and 95 countries that do not guarantee equal pay for equal work.[8]

Parity in educational opportunity and attainment is also a critical indicator of a country's resilience to catastrophic risk. Education empowers women, opening doors to employment, entrepreneurship, and leadership roles. Countries that invest in girls' and women's education see significant returns in terms of economic growth, health outcomes, and social cohesion. Educated women are more likely to participate in the workforce, contribute to innovation, and invest in their children's education, creating a virtuous cycle of development.

Figure 10: GCRI Score x Labor Force Participation Gender Divergence



Source: The World Bank

For example, countries like Germany, Sweden, and Canada have seen gains in competitiveness in large part due to their policies aimed at promoting women's education, workforce participation, and leadership roles. The integration of women in STEM (Science, Technology, Engineering, and Math) fields, in particular, has been instrumental in driving innovation.

Countries must also commit to enforcing domestic protections for women. These include equal rights within marriage and the ability to make autonomous decisions regarding finances and family life. In many countries, discriminatory marital laws limit women's autonomy over property, finances, and

8. World Bank. "Nearly 2.4 Billion Women Globally Don't Have Same Economic Rights as Men." World Bank, 1 Mar. 2022, www.worldbank.org/en/news/press-release/2022/03/01/nearly-2-4-billion-women-globally-don-t-have-same-economic-rights-as-men.

personal freedom. These legal inequalities often perpetuate a cycle of dependency, preventing women from pursuing opportunities for entrepreneurship or employment. In contrast, research has shown that families in which women have an equal say in household decisions, including financial decisions, experience better health outcomes, higher educational attainment for children, and greater economic security.

As explored in the Global Governance Forum's 2024/25 Gender Equality and Governance Index, the 20 worst-performing nations in terms of gender equality are also among the most violent countries in the world. This stark, yet unsurprising, correlation underscores the critical importance of advancing gender equality and implementing robust protections for women in order to enable national peace and stability. The correlation works both ways: when women are marginalized and lack such protections, societies often experience greater levels of violence and instability. Conversely, empowering women and ensuring their safety not only improves individual and community outcomes but also fosters more equitable governance and long-term societal resilience.

To illustrate the vast extent of progress that still needs to be made: 112 countries do not criminalize marital rape, 49 countries do not have specific laws against domestic violence, and 45 have no official legislation that address sexual harassment.[9]

Gender equality is more than just a moral imperative; it is a pragmatic necessity for any nation aspiring to mitigate an array of systemic risks while achieving stability and prosperity. Countries that recognize this and commit to phasing out the multiple human rights violations that are implicit in the confinement of women to the status of second-class citizens are better prepared to face the future.

9. World Bank. "SDG Atlas 2017: SDG 5 - Gender Equality." World Bank, 2017, datatopics.worldbank.org/sdgatlas/archive/2017/SDG-05-gender-equality.html.

CASE STUDY

WOMEN'S PARTICIPATION IN NIGERIA & RWANDA

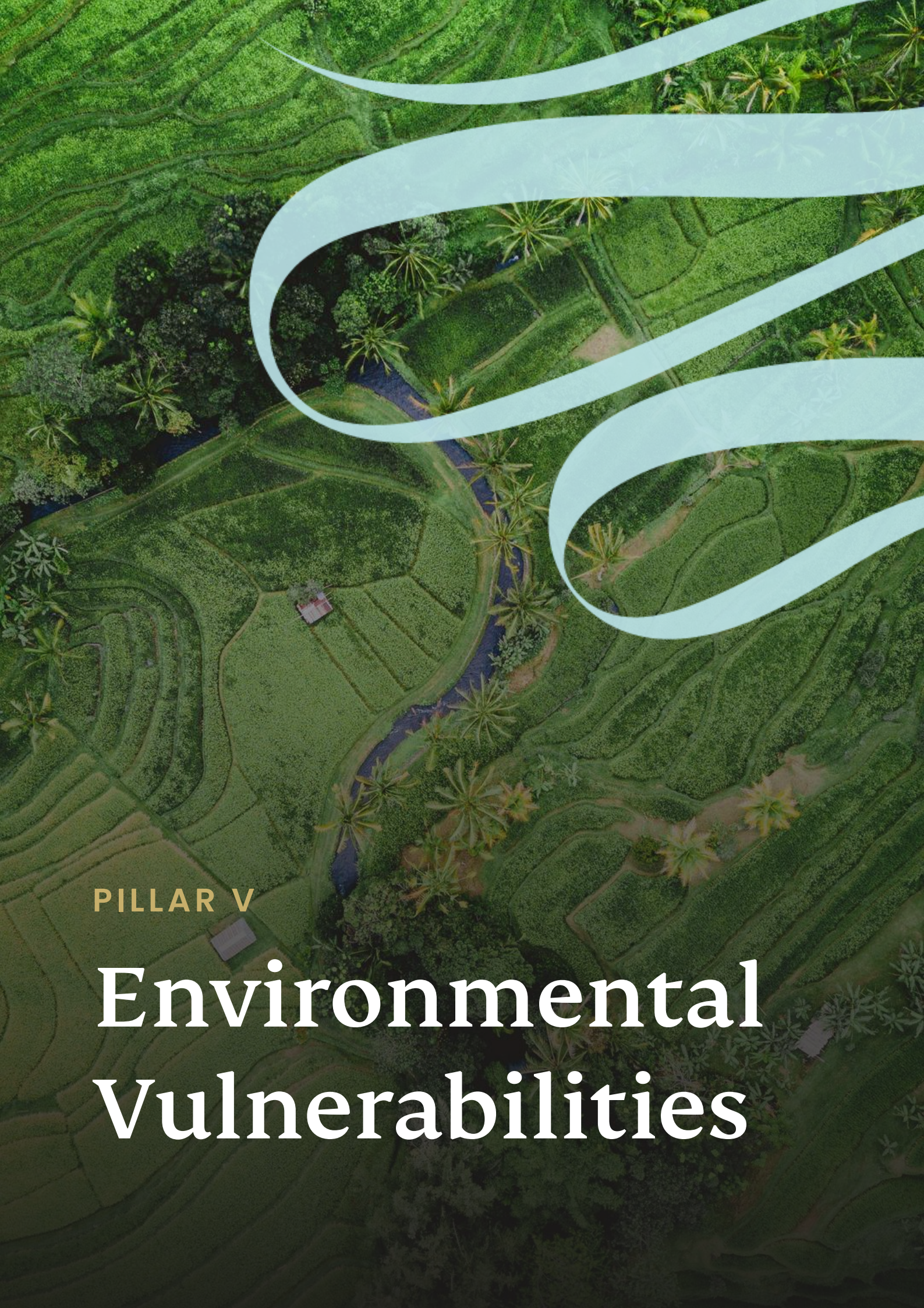
Women's participation in the political and economic spheres is crucial for a country's economic, political, and social stability, development, and vitality. Countries that prioritize gender inclusion typically experience more equitable economic growth, stronger governance, and enhanced social cohesion. Conversely, low female representation in leadership and the workforce is often linked to poorer development outcomes, greater vulnerability to internal conflict, and governance inefficiencies. Two African nations, Rwanda and Nigeria, provide a stark contrast in how differing levels of women's engagement can shape a nation's trajectory.

Rwanda is globally recognized for its high level of female political representation, with women holding 61% of parliamentary seats - the highest proportion in the world. This success is complemented by a relatively small gender disparity in labor force participation, with 86% of women participating compared to 88% of men. This remarkable achievement is the result of deliberate policy reforms following the 1994 genocide, which left women as the majority in the country. The Rwandan government adopted gender quotas and incorporated women's empowerment into the national development agenda, leading to the establishment of gender-sensitive policies and the integration of women in key decision-making roles.

The impact has been transformative. Women's inclusion has led to more responsive and equitable policies, focusing on healthcare, education, and community welfare. Economically, women are active in sectors beyond traditional roles, contributing to services and business. This broad-based participation has led Rwanda to be one of the fastest-growing economies in Africa, strengthening the country's resilience to economic shocks and lifting the standard of living. The high representation of women in governance has also been linked to lower levels of corruption and more effective public service delivery, bolstering the country's political stability.

In contrast, Nigeria's marginalization of women has hindered the development of the 4th-largest economy in Africa. Women hold only 4% of parliamentary seats and face a significant gender gap in labor force participation, with just 49% of women participating compared to 64% of men. Cultural barriers, discriminatory legal frameworks, and widespread gender-based violence perpetuate this exclusion, limiting women's ability to contribute economically and politically.

This exclusion has profound consequences. Nigeria's gender inequality stifles economic growth, with estimates suggesting that equalizing labor force participation rates could increase GDP by up to 23%. Politically, the low representation of women has resulted in policies that are less responsive to gender-specific issues such as maternal health, education, and community security. The exclusion of women from peacebuilding processes has also contributed to the country's persistent security challenges, particularly in conflict-prone regions where women's roles in conflict resolution and community stabilization are critical. As a result, Nigeria's lack of gender inclusivity exacerbates political instability and economic underperformance.



PILLAR V

Environmental Vulnerabilities

The recognition that environmental factors play a significant role in determining a country's catastrophic risk profile is on the rise. As climate change accelerates and its compounding risks are increasingly seen around the world, environmental measures have become critical indicators of a country's vulnerability to global catastrophic risk. Countries that recognize the importance of a sustainable environmental strategy and take proactive steps toward reducing their environmental footprint often enhance their resilience to climate-related risks. In contrast, countries that ignore these indicators and continue to degrade the environment face heightened exposure to both immediate and long-term threats, ranging from economic downturns to large-scale displacement of populations and geopolitical instability.

The global push for development to meet the needs of an expanding population has proceeded within the paradigm of linear supply chains and an economy maximizing profits, all while treating environmental and social impacts as externalities. The result has been the pillaging of a major part of Earth's land surface and of the natural resources with which the planet was once so abundantly endowed. Moreover, these efforts have generated pollution and waste on a massive scale. The oceans are similarly being stripped of resources, overwhelmed with plastics, acidified, and filled with our pollutants. To underscore the extent of damage that has been done, humanity is using the equivalent of 1.6 Earths to maintain our current way of life.[11] This has resulted in roughly 75% of the earth's land surface and 66% of the earth's ocean area being significantly altered by human activities.[12]

Our future is threatened as non-renewable resources become increasingly scarce and expensive, while normally, renewable resources are consumed beyond their rates of regeneration. To use an economic metaphor, we are living off the capital rather than the interest of our planetary biocapacity, with environmental footprints, especially of the affluent, far beyond what the Earth can sustain. Some of these changes are already irreversible, but most could be attenuated with rapid action toward both mitigation and adaptation in the next decade.

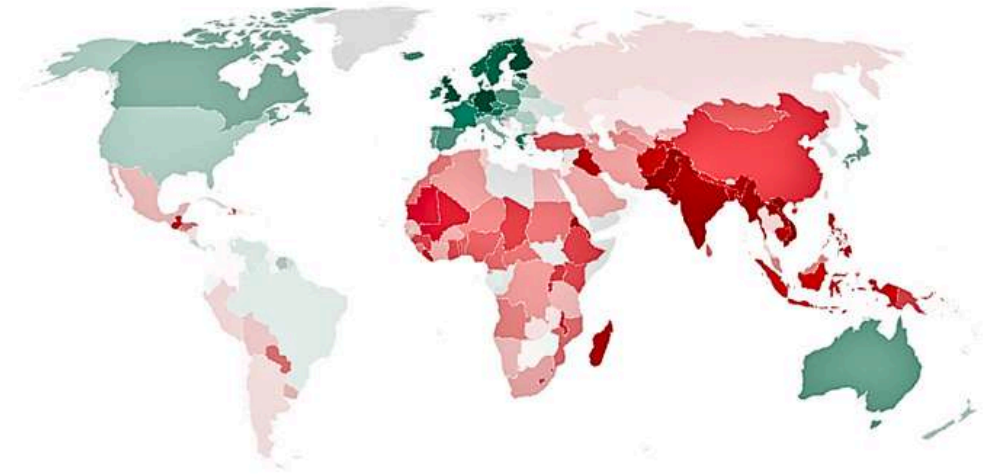
> **...we are living off the capital rather than the interest of our planetary biocapacity...**

Countries are responding to these challenges in varying ways. European nations like Germany, Denmark, and the Netherlands have adopted aggressive policies to reduce carbon emissions by transitioning to renewable energy sources and improving energy efficiency. In Asia, countries like South Korea and Japan are addressing air pollution by investing heavily in clean technology and reducing their reliance on fossil fuels. Response to climate change is not only a luxury reserved for high-income countries. Bangladesh, for example, which, due to its low-lying topography, is highly vulnerable to natural disasters, has implemented comprehensive climate adaptation strategies, including flood-resistant infrastructure and early-warning systems to mitigate damage from climate events. However, it remains true that countries in the developing world struggle disproportionately with environmental risk. Unfortunately, it is these countries that lack the financial or technological resources to adapt to climate change that are at a greater risk of experiencing its most devastating effects.

11. Food and Agriculture Organization of the United Nations. The Conservation of Healthy Ecosystems and Biodiversity for Livelihoods and Food Security. FAO, openknowledge.fao.org/server/api/core/bitstreams/4e57915f-a5a3-4867-a771-57cceb36230b/content.

12. Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). Summary for Policymakers of the Thematic Assessment of the Sustainable Use of Wild Species. IPBES, files.ipbes.net/ipbes-web-prod-public-files/downloads/spm_unedited_advance_for_posting_htn.pdf.

Figure 11: Environmental Performance Index Score

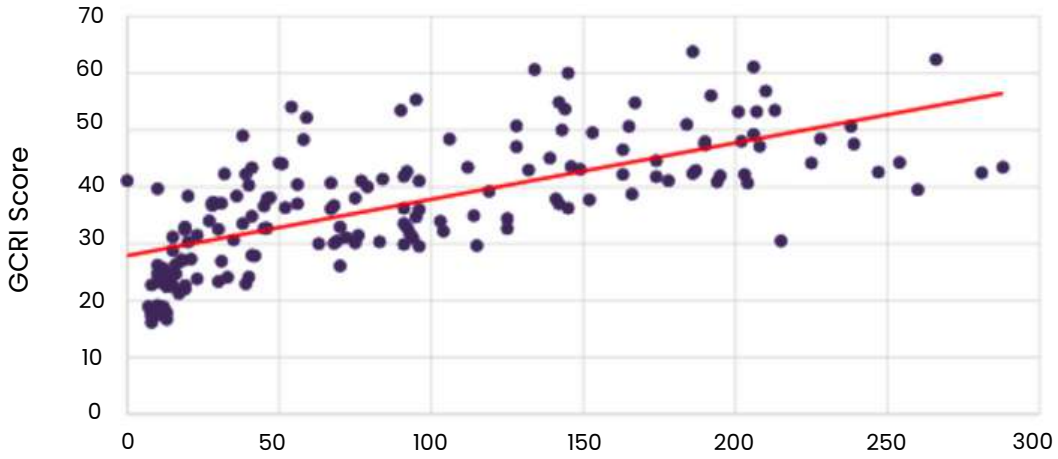


Source: Yale Center for Environmental Law & Policy

Air pollution is a critical environmental risk indicator that has significant consequences for a country's stability and development. Nations with high levels of air pollution face clear risks such as increased public health costs, but also face less obvious risks including reduced workforce productivity. It is estimated that exposure to air pollution reduced average global life expectancy by 2.2 years, an effect that is comparable to that of smoking, three times more detrimental than unsafe water, and 89 times more detrimental to average life expectancy than conflict and terrorism.[13] In countries like China and India, air pollution has reached truly alarming levels, leading to widespread health issues and premature deaths. To combat this, cities such as Beijing and New Delhi have implemented air quality control measures, including restrictions on vehicle emissions and the closure of coal plants. These policies are well-intentioned, but the magnitude of their impact has yet to match the magnitude of the crisis.

13. World Bank. "Nearly 2.4 Billion Women Globally Don't Have Same Economic Rights as Men." World Bank, 1 Mar. 2022, www.worldbank.org/en/news/press-release/2022/03/01/nearly-2-4-billion-women-globally-don-t-have-same-economic-rights-as-men.

Figure 12: GCRI Score x Pollution Deaths



Source: The World Bank

Rising global temperatures are exacerbating the frequency and severity of natural disasters, including hurricanes, floods, wildfires, and droughts. Countries that are unprepared for these climate-induced events are particularly vulnerable to global catastrophic risks. Droughts, for instance, can cripple agricultural economies, cause food shortages, and trigger mass migration, while coastal storms and flooding can devastate infrastructure and displace millions of people. With sea levels expected to rise 12-14 inches by 2050, hundreds of millions of people could be displaced globally.[14]

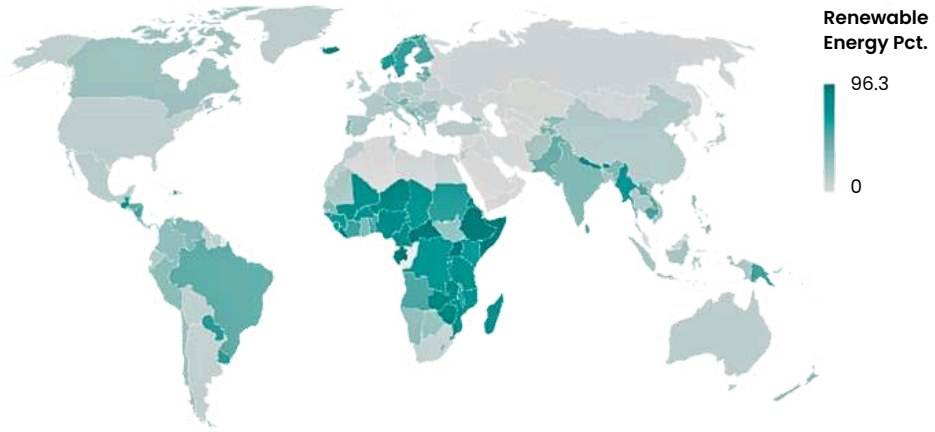
One often overlooked environmental factor that contributes to a country's vulnerability is pesticide use. The overuse of harmful pesticides can degrade soil health, reduce biodiversity, and contaminate water sources. This not only threatens agricultural productivity but also endangers public health. For countries dependent on agriculture, high pesticide usage exacerbates their vulnerability to global food security risks, particularly as climate change stresses agricultural yields.

Countries such as Costa Rica and India have launched initiatives to reduce pesticide use by promoting organic farming and adopting integrated pest management strategies, which help preserve the environment and reduce health risks to the population. These approaches not only improve food security but also contribute to sustainable economic development by preserving ecosystems that underpin agricultural productivity.

The shift toward renewable energy is one of the most important strategies in mitigating climate risk. Countries that rely heavily on fossil fuels are not only contributing to global warming but also increasing their exposure to energy market volatility and supply disruptions. Renewable energy sources — including solar, wind, and hydropower — offer sustainable energy that is also less prone to supply and pricing fluctuations.

14. National Oceanic and Atmospheric Administration. "U.S. Coastline to See Up to a Foot of Sea Level Rise by 2050." NOAA, 15 Feb. 2022, https://www.noaa.gov/news-release/us-coastline-to-see-up-to-foot-of-sea-level-rise-by-2050.

Figure 13: Renewable Energy (% of Total Energy Consumption)



Countries like Iceland and Costa Rica have already achieved near-complete renewable energy sourcing (Iceland is at nearly 100% and Costa Rica is at 98% renewable-sourced energy), relying primarily on geothermal and hydropower, respectively. Their commitment to clean energy has insulated them from the geopolitical risks associated with fossil fuel dependence while promoting long-term economic stability. In contrast, countries that continue to rely on fossil fuels — such as Saudi Arabia (99% of total energy consumption), South Africa (94%), Thailand (92%) and Mexico (90%) — face greater risks from both economic and environmental shocks.[15]

15. Energy Institute - Statistical Review of World Energy (2024).

CASE STUDY

AIR POLLUTION IN THE PHILIPPINES & THAILAND

Ambient air pollution presents risks to all members of a population but disproportionately affects those with pre-existing conditions or other at-risk populations. Although pollution is often a product of industrial activity, it often cycles back as a drain on national economies when workers are impacted and their productivity is disrupted. The impact of disparate pollution levels can be seen in a comparison of two Southeast Asian countries, the Philippines and Thailand.

The Philippines suffers from one of the highest rates of death attributable to ambient air pollution in the world, with 203 deaths per 100,000 citizens. The country's rapid urbanization and industrialization have contributed to high levels of pollution, particularly in urban centers. The country's air pollution problem is exacerbated by a lack of stringent environmental regulations and enforcement. While there are several laws in place to combat pollution, enforcement remains weak concerning industrial and motor vehicle emission controls. Moreover, the country's heavy reliance on fossil fuels, including coal, for electricity generation further worsens the country's air quality. This reliance on outdated energy sources, coupled with ineffective policy implementation and weak enforcement, has led to persistently high pollution levels.

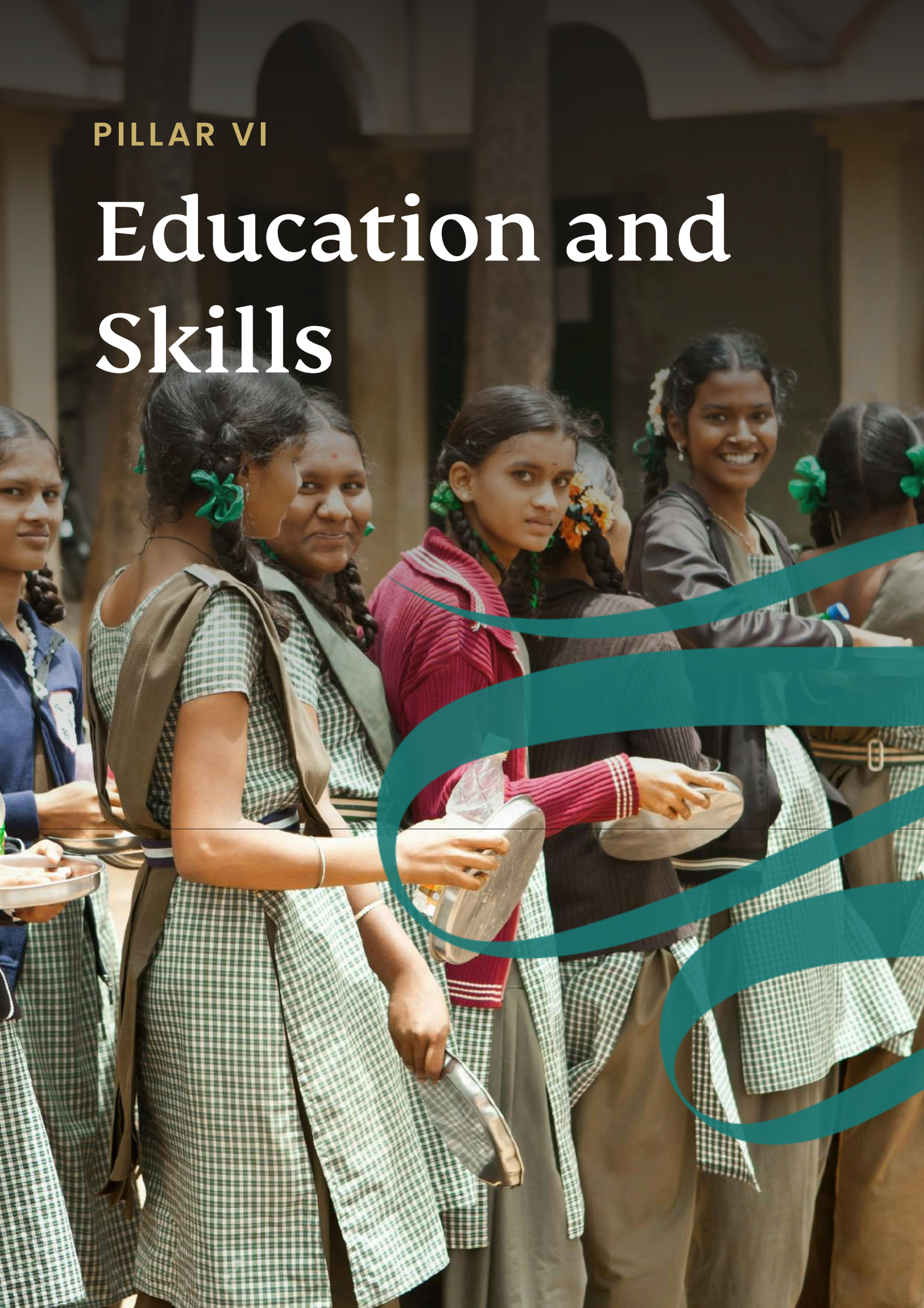
High pollution levels in the Philippines have had a critical impact on public health, particularly among vulnerable groups such as children, the elderly, and those with pre-existing respiratory conditions. Beyond a substantial toll on human well-being, the economic burden of this is substantial, as the country spends heavily on healthcare costs related to pollution-induced illnesses. Additionally, the loss of productivity due to air pollution-related health issues, including absenteeism from work and long-term disability, hinders the country's economic prospects.

In contrast, Thailand has made notable progress in addressing air pollution, resulting in significantly lower levels of death attributable to ambient air pollution (46 deaths per 100,000 citizens). Despite rapid industrial growth and urbanization, Thailand has taken a more active approach to managing air pollution. For instance, it has implemented a series of robust environmental policies and closely regulates industrial emissions. The country has also invested heavily in promoting clean energy, which has reduced the country's reliance on coal and other polluting fuels. Motor vehicle emissions standards and incentives for electric vehicle adoption have further contributed to the overall reduction of air pollution, particularly in urban areas like Bangkok.

The health dividend of these policies has been significant. Lower levels of air pollution have resulted in reduced respiratory and cardiovascular disease, translating to higher economic productivity as fewer workers are affected by pollution-related health issues. In addition, Thailand's relatively cleaner air has had a positive impact on attracting tourism, a key driver of the country's economy.

PILLAR VI

Education and Skills



Education is a cornerstone for economic growth, social stability, and political development. At the individual level, education is a fundamental enabler for citizens' effective participation in the economic and political life of their countries and local communities. Quality education not only spurs individual development but also drives economic growth by equipping citizens with the skills necessary to participate in an increasingly globalized and technology-driven economy. For developing nations, effective education can mean the difference between being trapped in a cycle of poverty and economic stagnation or becoming a thriving, globally competitive economy.

Effective education can mean the difference between being trapped in a cycle of poverty and economic stagnation or becoming a thriving, competitive economy.

Conversely, a lack of basic information and skills increases risk by severely limiting citizens' ability to participate in the development process, to be gainfully employed, to be well-informed judges of government policies, and to avoid falling prey to the manipulations of demagogues. Large segments of a poorly educated population often struggle to find employment or participate meaningfully in their economies, which can fuel unrest, populist movements, and extremism.

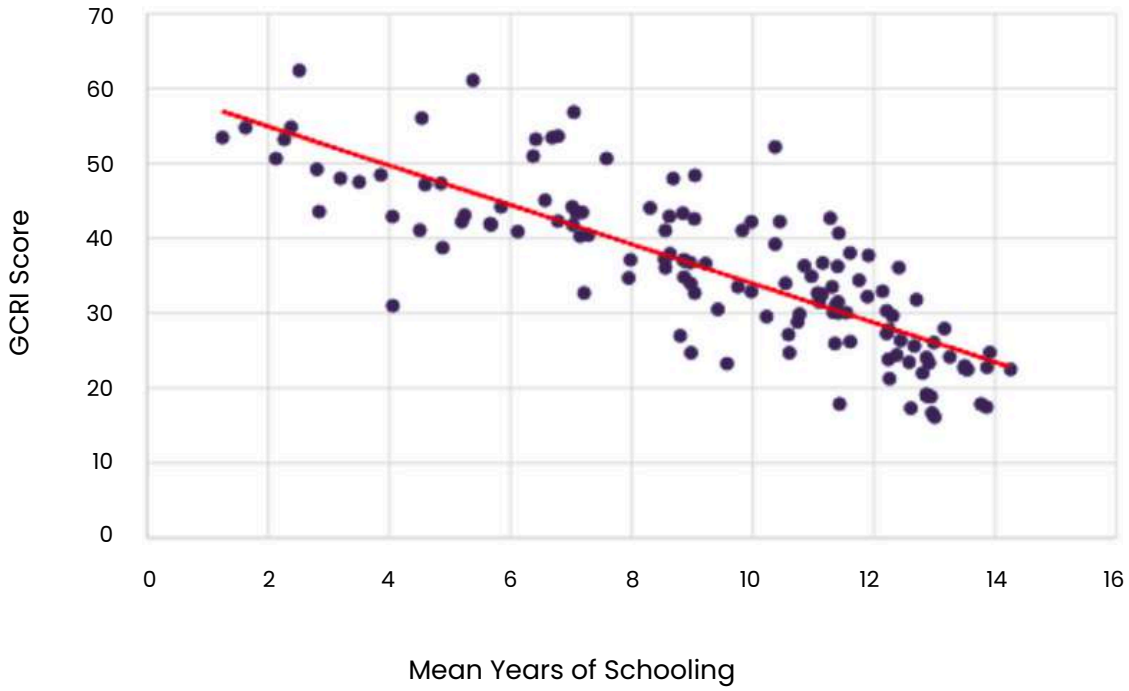
Figure 14: Quality of Governance Pillar Score x Education and Skills Pillar Score



Recent decades have seen a global emphasis on improving educational outcomes – as of 2023, adult literacy rates had reached 87%, up from 82% two decades earlier.[16] School enrollment rates, particularly in primary education, have also shown impressive gains, with many countries achieving near-universal access to basic education. According to UNESCO, nearly 90% of children worldwide are now enrolled in primary school. These aggregated global figures are encouraging, but the GCRI's country-level analysis reveals striking disparities in the quantity and quality of education, access to vocational training, and brain-drain effects facing countries.

16. The World Bank.

Figure 15: GCRI Score x Quantity of Education



Source: UNESCO

Regions with high levels of political instability and gender inequality, such as Sub-Saharan Africa and parts of the Middle East, often face severe disruptions to their education systems. This creates a self-perpetuating cycle wherein widespread illiteracy, and a lack of vocational training opportunities contribute to heightened social and economic vulnerability. On the other hand, countries with more robust and inclusive education systems, such as those in Scandinavia or East Asia, consistently rank higher in global competitiveness, exhibit lower levels of unemployment, and experience greater political stability.

Despite the encroachments of automation and AI, human capital will remain a critical endowment and indicator of relative competitiveness in the global markets of today and the future. Workers must have access to new knowledge, including continual training in new processes and in the operation of the latest technologies. As automation and AI continue to reshape the workforce, countries with well-educated and highly skilled populations will be better positioned to navigate future challenges, adapt to new trends and technologies, and sustain long-term prosperity.

Additionally, countries that emphasize vocational training tend to be more resilient to risks, including economic shocks. For example, Germany's dual education system, which combines traditional classroom-based learning with apprenticeships in industries such as manufacturing and engineering, has helped the country establish one of the world's lowest unemployment rates. On the contrary, countries that fail to offer vocational training often see higher unemployment rates, especially among young people, which can in turn contribute to political and social instability.

Both the benefits and risks associated with a country's education system are magnified by the gender parity in that system. Globally, girls and women continue to face barriers to education, including cultural and religious norms. Gender disparities in education have profound implications, not just for individual

women but for entire economies. For instance, it has been estimated that 10% more adolescent girls attending school would increase a country's GDP by an average of 3%.[17]

Educating girls also leads to lower fertility rates, improved child health, and greater participation in the workforce. Women who receive education are more likely to start businesses, invest in their children's education, and contribute to their communities in a range of other ways. Rwanda, for instance, is an example of a success story in this regard and has made large strides in closing its gender gap in education, resulting in one of the highest female labor force participation rates in Africa.

Even when countries succeed in the education of their populace, many countries still struggle to overcome brain drain — the emigration of high-skilled workers, typically to more advanced economies that present greater financial opportunity. When countries invest in educating their citizens but fail to provide adequate economic opportunities, they risk losing their most talented individuals to countries with better prospects. Past and present migration fuels future migration by exacerbating existing inequalities and hampering the country's ability to develop its economy.

Brain drain is most effectively combatted through the establishment of political stability and economic development. However, some developing countries have also instituted policies to combat brain drain. Some have done this by strengthening ties with their diaspora communities. Ghana and India have both done this to great effect. Others, such as Chile, have created incentives for skilled workers to return home. Chile's Start-Up Chile program provides equity-free grants and visas to entrepreneurs, encouraging them to start businesses in Chile.

The establishment and maintenance of an effective education system is fundamental to the achievement of a country's development aims. Education, more than many other indicators, is intricately interwoven with almost every risk factor facing a country, and succeeding in education can yield substantial benefits across a country's resilience profile. Investing in education and training also helps governments enhance the quality of their civil services, in turn augmenting their ability to formulate effective public policies, particularly in areas that help mitigate various risk factors.

17. USAID.

CASE STUDY

EDUCATIONAL ATTAINMENT IN BOTSWANA & NIGER

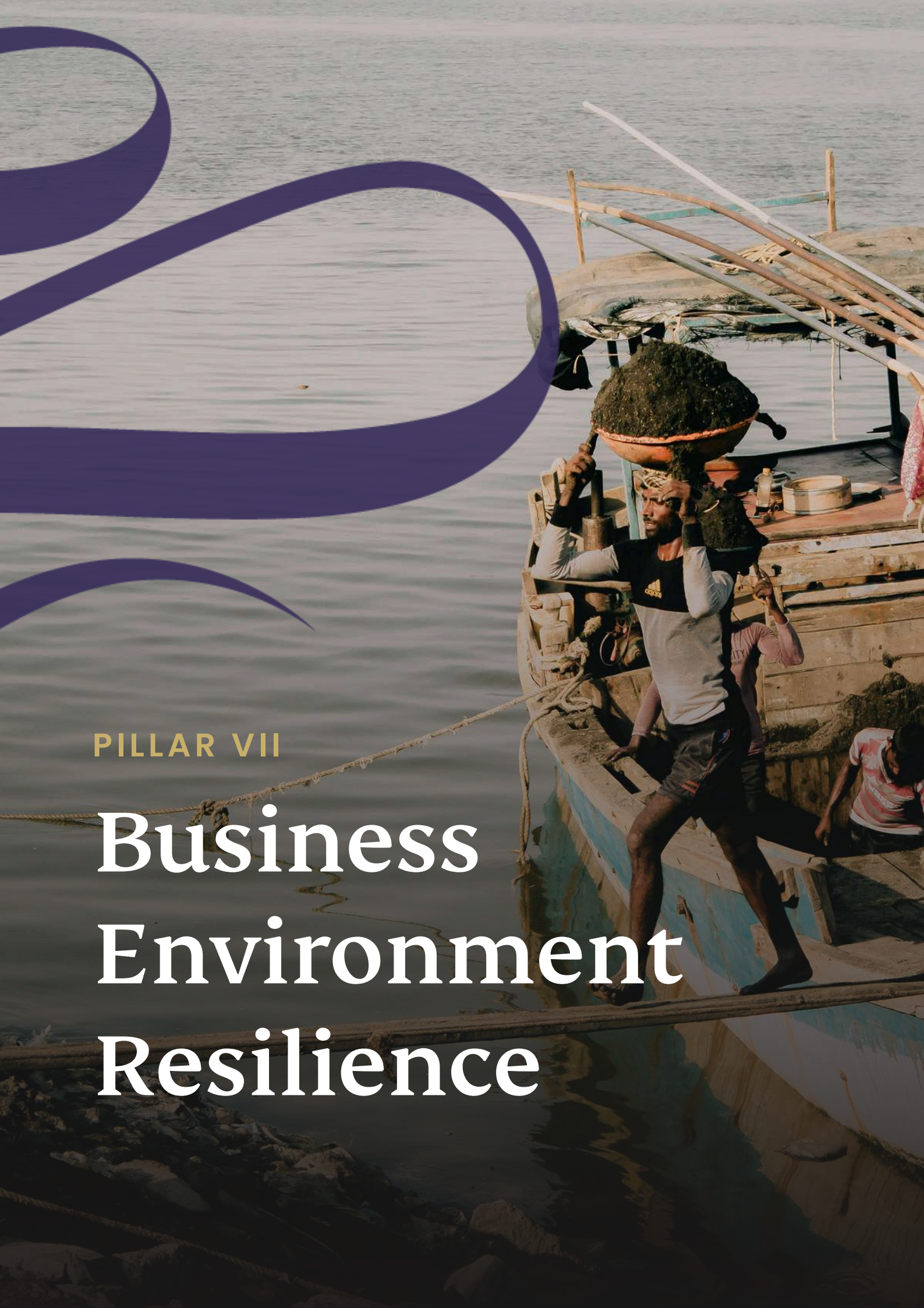
Education plays a critical role in shaping a country's economy, human capital endowment, capacity for innovation, and overall stability. Countries with high levels of educational attainment tend to have more economic resilience, a greater ability to innovate, and lower vulnerability to domestic and international risks. In contrast, countries with lower scholastic achievement are more vulnerable to instability and tend to rank higher on global risk indices. This can be seen through a comparison of two African countries, Niger and Botswana.

Niger, with an average of just 1.6 years of schooling and a primary school enrollment rate of only 65%, faces a deeply entrenched educational crisis that perpetuates a cycle of poverty by limiting economic opportunity. The country's adult literacy rate remains low at 38%, hindering its ability to cultivate a skilled workforce. Consequently, Niger's economy is driven by low value-add sectors such as agriculture (which employs nearly 85% of its labor force), small-scale mining, and low-tech industry, creating limited opportunities for social mobility and economic diversification.[18] These reduced economic prospects have fostered political instability and conflict, as low levels of education weaken civil society's capacity to engage in governance and hold leaders accountable.

In contrast, Botswana has achieved a 94% primary school enrollment rate, supported by spending 8% of its GDP on education—one of the highest rates in Africa and more than double Niger's expenditure. This substantial investment has translated into an adult literacy rate of 87% and a more skilled labor force capable of supporting high-value sectors such as finance, tourism, and higher-tech manufacturing. The country's education strategy, guided by initiatives such as the Education and Training Sector Strategic Plan, focuses on expanding access, improving teacher training, and strengthening technical and vocational education. As a result, Botswana's economy has moved beyond raw resource exports to include activities like diamond processing, gem cutting, and tourism, making it less vulnerable to commodity price fluctuations and more resilient to global economic cycles.

Ultimately, the contrasting cases of Niger and Botswana illustrate how strategic investments in education can shape a country's economic trajectory and social stability. While Niger's underdeveloped education system contributes to a vicious cycle of poverty and conflict, Botswana's focus on education has enabled it to build a more diversified, high-value economy and achieve sustained growth.

18. Feed the Future. "Niger." Feed the Future, U.S. Government's Global Hunger and Food Security Initiative, https://www.feedthefuture.gov/country/niger/.



PILLAR VII

Business Environment Resilience

Pillar VII: Business Environment Resilience

The quality of a country's business environment reflects not only the ease with which individuals and firms can enter and operate in the national economy but also indicates the extent to which state institutions are functioning effectively. Like the businesses that operate in it, a national business environment thrives on efficiency, predictability, and established rules. Property rights, an independent judiciary, stable regulations, and a transparent tax regime can work in tandem to create an inclusive and level playing field. This in turn promotes investment and entrepreneurship. However, in the absence of these components, businesses are less able to plan for the future and, as a result, are less likely to innovate, hire, and invest in their future. In some cases, individuals and enterprises may be forced into the underground economy or be inhibited from starting in the first place.

Like the businesses that operate in it, a national business environment thrives on efficiency and established rules.

Entrepreneurship is a critical driver of job creation, innovation, and development. Countries that have high levels of entrepreneurial activity are more likely to experience sustained economic growth and to adapt quickly to new global challenges. In addition, countries with fertile entrepreneurial landscapes are often at the forefront of technological advancements. Facilitating entrepreneurship through supportive policies—such as reducing regulatory burdens, providing access to credit, and offering incentives for business formation—helps unlock the full potential of the private sector.

Moreover, entrepreneurialism is a critical pathway to individual empowerment, conferring opportunities for individuals to take control of their economic destiny, become more self-reliant, improve their standard of living, and break free from cycles of poverty. Greater financial independence at the individual level confers benefits at the community level as well, increasing investment in education, health, and community development, in addition to fostering upward mobility and greater long-term stability.

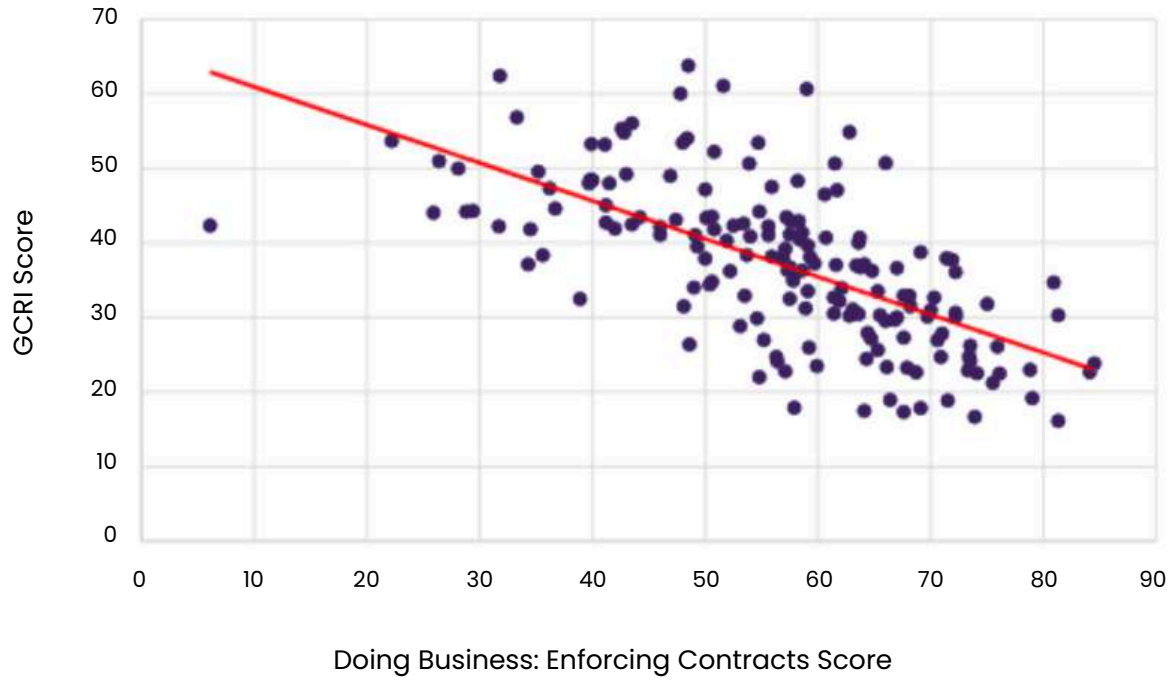
Conversely, in countries where entrepreneurship is stifled, risk multiplies as the private sector's ability to contribute to growth is restricted, unemployment rises, economic growth stagnates, and the country becomes more vulnerable to global shocks. A dearth of entrepreneurial opportunity leaves individuals entrenched in systems that may include cyclical poverty, precarious employment, and an outflow of human capital to countries offering the prospect of greater opportunity.

Turning to specific business environment indicators, property rights are one of the fundamental guarantees that a state must underwrite for its citizens. They determine the terms of peaceful, transparent, and inclusive market competition for scarce resources. When individuals and businesses have confidence that their assets—whether physical property, intellectual property, or financial assets—are secure, they are far more likely to invest, innovate, and participate in the economy. Secure property rights create incentives for investment in technology, equipment, and human capital, driving economic productivity and fostering long-term development. An effective regime in this regard gives individuals and businesses they have the exclusive right to employ their resources as they see fit and are protected from expropriation.

In contrast, a weak property rights regime heightens risk by creating uncertainty, discouraging investment, and potentially leading to unrest. In countries where property rights are not well-protected, there is often widespread corruption and expropriation, which in turn undermines trust throughout the country. Among the twenty countries with the weakest property rights regimes, the average scaled Corruption Perceptions Index score is 0.66. Among the twenty countries with the strongest property rights regimes, the average scaled score is 0.10. Fear of corruption and expropriation, resulting in reduced economic activity, makes countries more vulnerable to risks such as capital flight, economic collapse, and political instability.

Similar to property rights is the concept of contract enforceability – the extent to which legal agreements are binding and enforced such that a party is held accountable for violating the terms of an agreement. Contract enforceability is a bedrock of a quality business environment and a key component of a strong rule of law. When businesses and individuals have confidence that contracts will be upheld by the legal system, they are more likely to engage in long-term economic activities such as trade, investment, and partnership. Contract enforceability builds trust, reduces transaction costs, and engenders predictability.

Figure 16: GCRI Score x Contract Enforceability



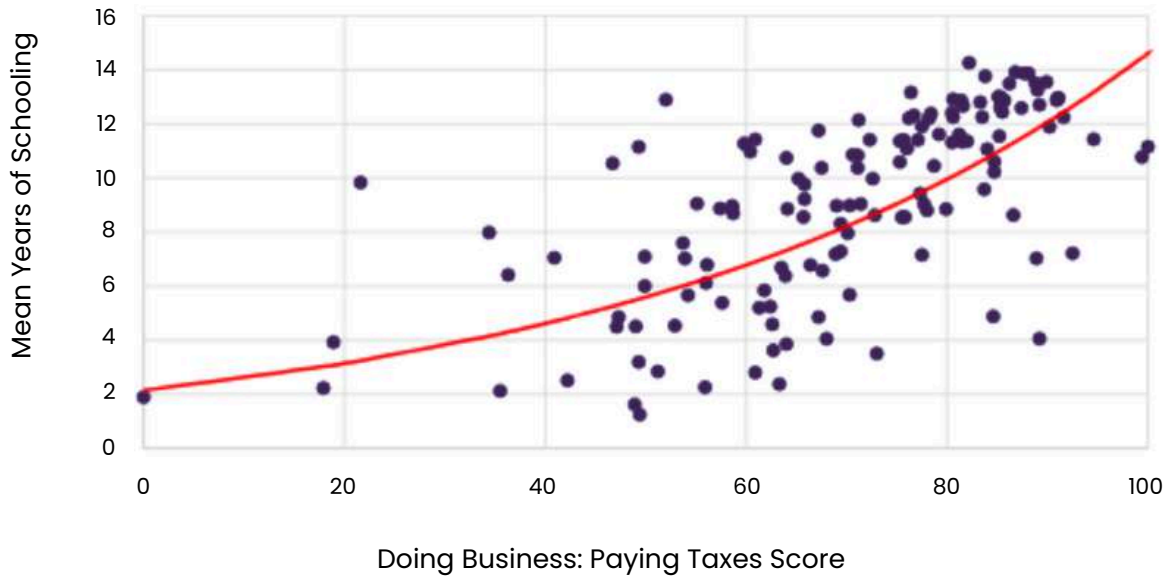
Source: The World Bank

Regulation is the major avenue through which governments can influence the business environment of their country. However, regulations must be transparent, efficient, and equitable to foster, rather than restrict, entrepreneurship and investment. Unfair or excessively burdensome regulation can hamper growth and encourage underground economies. An excessively burdensome business environment, with arbitrary rules that serve as vehicles for bureaucratic discretion and corruption, will have adverse economic effects. Risk will grow in tandem with higher levels of unemployment and lower levels of revenue collection, both of which impair the ability of the state to respond to vital social needs and instead foster social discontent, political instability, and heightened risk.

Tax collection is another pillar of the modern state's mandate. Efficient and effective tax collection provides the state with the resources it requires to perform its duties. An effective tax regime can discourage evasion and the resulting growth of the underground economy. Effective and equitable tax collection systems are essential for funding public services, infrastructure, and social programs that promote stability and development. When countries can efficiently collect taxes, they can invest in critical areas like education, healthcare, and transportation that yield benefits to the country more broadly. In practice, the extent to which collected taxes are recycled into the economy and into meaningful social services is striking: among the 50 most effective tax-collecting countries, mean years of schooling

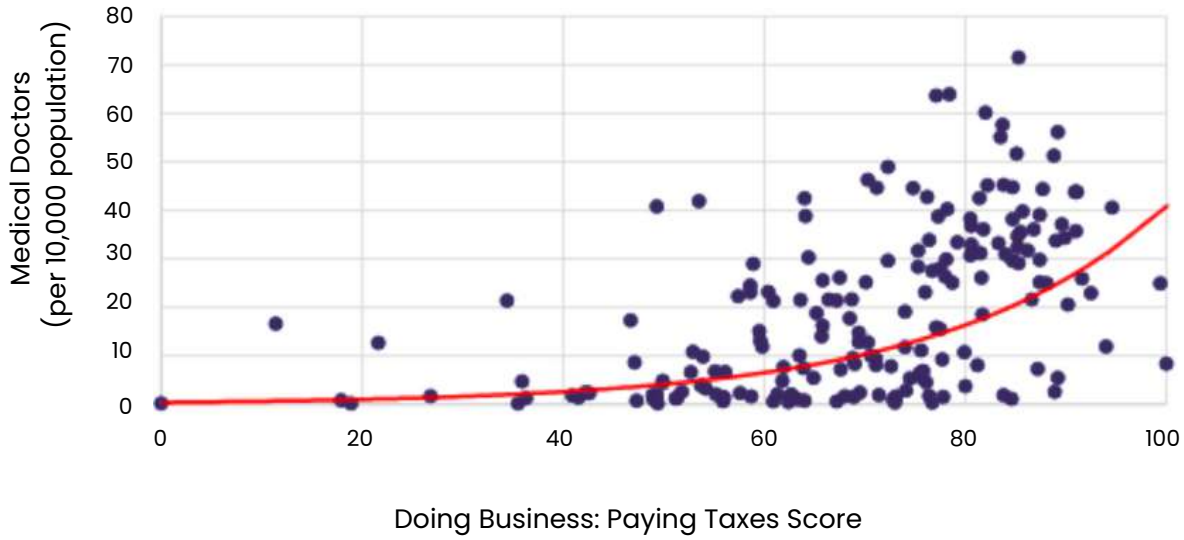
averages 11.7 and doctors per 10,000 population averages 32.9. In the 50 least effective tax-collecting countries, mean years of schooling averages 6.3 and doctors per 10,000 population averages 8.5. Strong tax collection systems also enhance fiscal transparency and reduce opportunities for corruption, which is vital for maintaining public trust and attracting international investment. Countries that fail to enforce tax laws or struggle with rampant tax evasion often face chronic budget deficits, leading to reduced public services and increased debt.

Figure 17: Tax Collection x Education



Source: The World Bank

Figure 18: Tax Collection x Healthcare



Source: The World Bank

CASE STUDY

COMMERCIAL DISPUTE RESOLUTION IN BANGLADESH & SINGAPORE

The efficient and transparent resolution of commercial disputes is a prerequisite for the functioning of a nation's economy, business environment, and marketplace. Reliable dispute resolution mechanisms enable trust, facilitate investment, and bolster business confidence. In the absence of a capacity to resolve commercial disputes effectively, countries suffer, and the disparity is on full display comparing two Asian countries — Singapore and Bangladesh.

With a score of 87.07, Singapore is among the highest-scoring countries on the Ease of Resolving a Commercial Dispute indicator within the World Bank's Business Ready report. The high score is the result of an independent, transparent, and efficient judicial system that is underpinned by a robust rule of law. In addition, Singapore's judicial system includes specialized commercial courts (the Singapore International Commercial Court), alternative dispute resolution mechanisms, and digital case management systems to streamline the resolution of commercial disputes. This legal infrastructure has yielded an average time to resolve a contract dispute of 164 days. While nearly 6 months may seem a long time, it is nearly 75% below the global average of 650 days (or nearly 22 months). This efficiency enhances business confidence and reduces transaction costs, resulting in a thriving financial services sector and sustained economic growth as companies can plan and execute long-term strategies without fear of counterparties not being held accountable for violating the terms of commercial agreements.

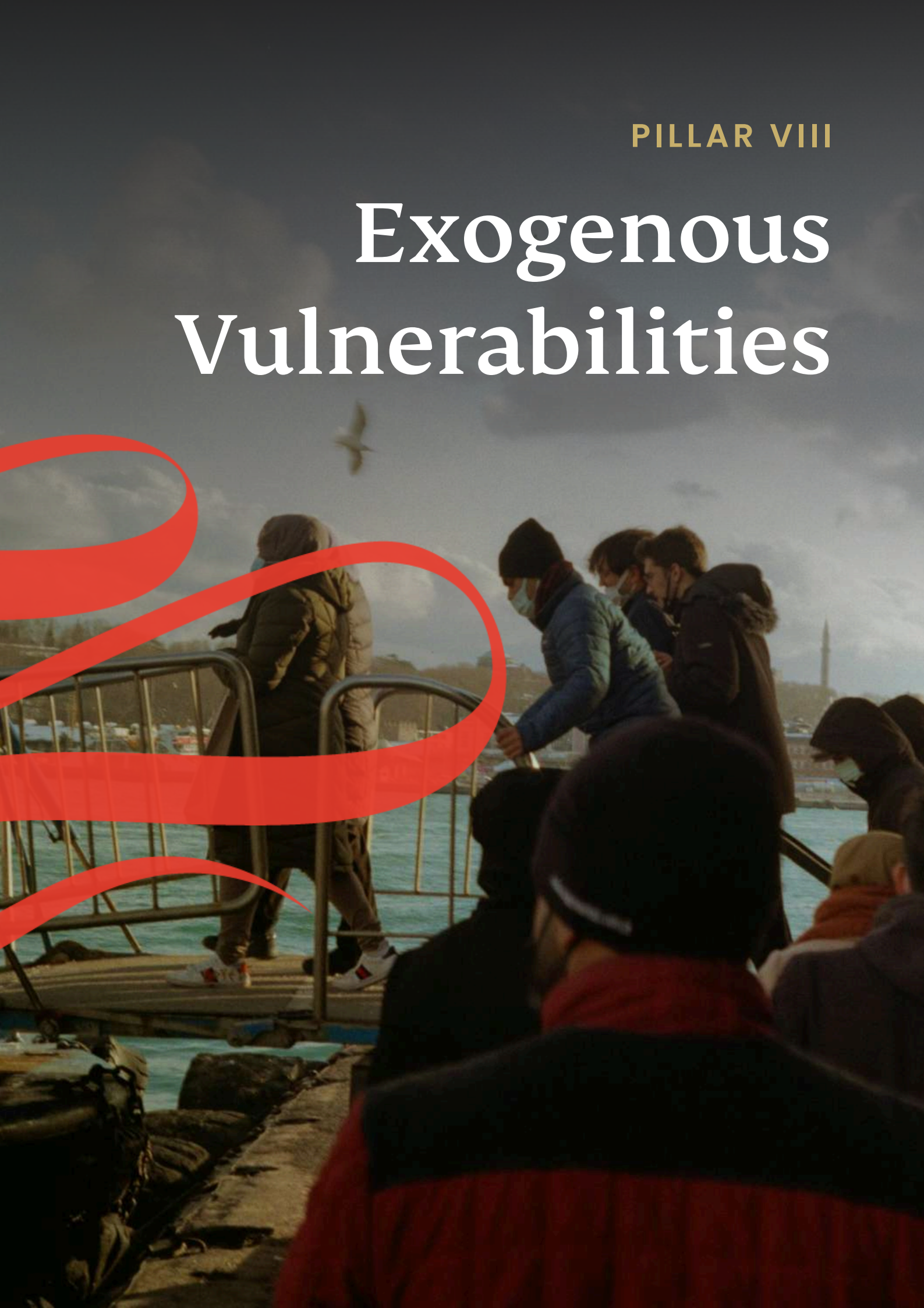
In stark contrast, Bangladesh's commercial dispute resolution score of just 56.33 is one of the lowest globally. Owing to a severe backlog of cases, judicial corruption, and a lack of specialized courts for commercial disputes, contract enforcement in Bangladesh can take up to 1,442 days—almost nine times longer than in Singapore and more than double the global average. Moreover, the cost of enforcing a contract can amount to 67% of the claim value, compared to just 26% in Singapore. This weak commercial legal environment creates an environment of uncertainty and mistrust among both domestic and foreign businesses operating in Bangladesh.

Additionally, the lack of timely or effective dispute resolution undermines the development of a formal credit market in the country. Financial institutions are reluctant to extend credit to businesses without reliable collateral enforcement mechanisms, limiting access to finance and stifling business expansion. Together, these factors discourage both local entrepreneurship and foreign investment. As a result, many enterprises in Bangladesh remain informal, unable to scale up or integrate into the global value chain. The inability to resolve disputes quickly also affects labor markets, as businesses may hesitate to engage in formal employment contracts, perpetuating informal labor practices and disadvantaging workers through an absence of labor protections and slower wage growth.



PILLAR VIII

Exogenous Vulnerabilities



The preceding pillars focus on thematically grouped indicators. Although not stand-alone pillars, the following risk indicators grouped under the subheadings of demographics, technology, and defense are nonetheless crucial considerations for the development of an accurate risk index. In addition, they are often both causal and symptomatic of the indicators explored in the Index's prior pillars.

Demographics

Swelling populations within fixed borders have resulted in significantly increased population densities in most countries over the past thirty years. In densely populated areas, economic activity tends to be concentrated, driving innovation, facilitating infrastructure development, and supporting diverse economic activities. However, high population density has also led to risks including a strain on resources, increased congestion, and damage to the environment. Pollution, loss of habitat, stress on clean water, and a reliance on pesticides and steroids to augment the food supply are all examples of increased population density's negative impacts.

The risks of high population densities are most pronounced in low-elevation coastal areas. Such areas are already at an increased risk of flooding and climate change will only exacerbate this risk. As noted previously in this report, rising sea levels will force migration away from inundated coastal lands, and the resulting increase in saline and brackish water bodies will increase the transmission risk of many water-borne infectious diseases such as cholera, typhoid, and hepatitis.

A driver of population density, migration is a global phenomenon and a process with far-reaching implications for origin and destination countries alike. Worldwide, the number of migrants continues to grow rapidly, reaching 281 million - 3.6% of the global population - according to UN figures.[19] Although immigration contributes positively to inclusive growth and sustainable development, it remains a controversial political subject in countries around the world. Native populations often fear competition for finite employment opportunities, a depression in salaries at the lower end of the income distribution, free-riding on public resources, and a perceived dilution of their culture. With respect to the allegation of free-riding, research has shown that even low-educated migrants have a better public resource footprint – the difference between their contributions and the benefits they receive – than native-born peers. However, regulating immigration is rarely a simple task, and many migrants enter countries illegally. This increases the need to allocate border and police expenditures away from other more productive ends.

> **Like the businesses that operate in it, a national business environment thrives on efficiency and established rules.**

Technology

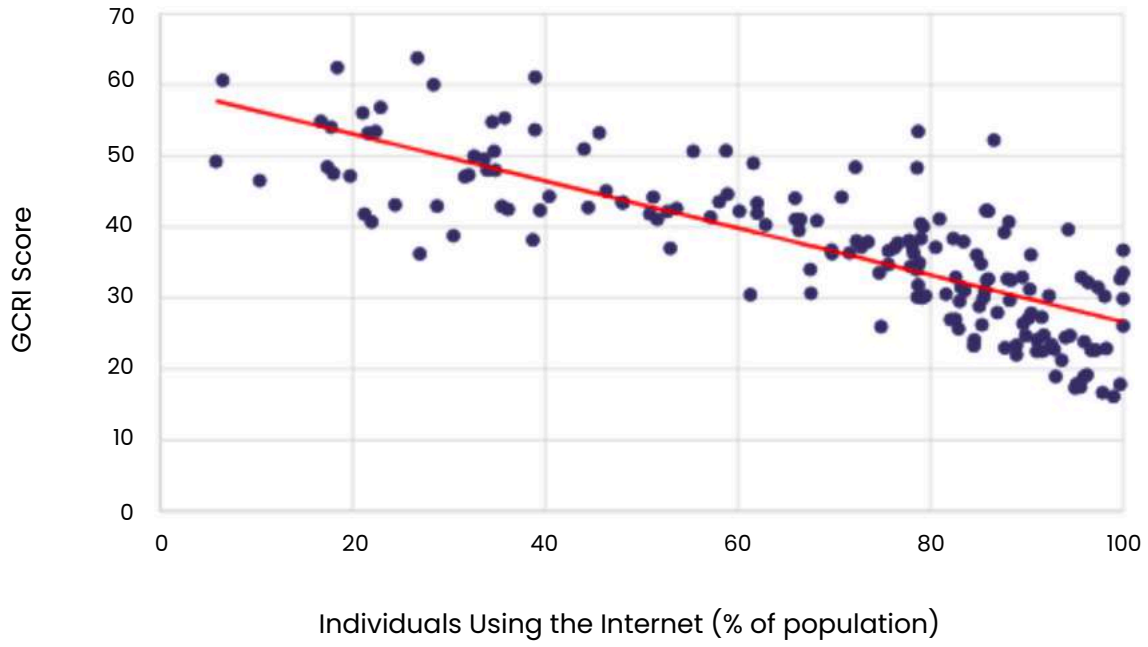
In the tech-driven global economy, internet access and cellular connectivity are prerequisites to full participation. Countries with high internet penetration rates tend to be better positioned to foster economic growth, promote innovation, and improve governance transparency. In countries with limited internet access, large segments of the population risk being excluded from critical information and economic opportunities, thus increasing inequality. Connectivity is not an unqualified good, however: countries with high internet penetration must contend with the risks of cyberattack, misinformation, and digital surveillance, which can undermine political stability and national security.

19. UN Migration. World Migration Report 2024.

Similarly, the rise of artificial intelligence and machine learning present opportunities, but also significant risks as their development outpaces regulatory and ethical safeguards. AI technologies' collection of personal data often proceeds without consent, enabling state surveillance at an unprecedented scale. Furthermore, workforce displacement poses a profound challenge for all countries (particularly those in the most tech-enabled economies) with automation threatening to displace hundreds of millions of jobs across industries. This displacement has the potential to push income inequality and social unrest to new heights globally. Perhaps most alarming is the misuse of AI, from the deployment of autonomous weapons to the proliferation of misinformation, which can destabilize societies and undermine trust in institutions. Coordinated global oversight and regulation efforts are urgently needed to manage these risks. In their absence, an unchecked rise of AI will almost certainly lead to economic, social, and political crisis.

Across the world, particularly in rural or remote areas, mobile networks are the primary means of communication, enabling access to financial services, healthcare, and education. Countries with extensive cellular coverage are better able to communicate with their populations during emergencies, facilitate disaster response, and coordinate relief efforts. In regions with poor mobile network infrastructure, such as parts of sub-Saharan Africa and Southeast Asia, lack of connectivity increases risk by hindering economic development, limiting access to services, and increasing vulnerability to security threats. Expanding cellular networks and improving coverage is essential for enhancing resilience, particularly in regions prone to natural disasters or conflict.

Figure 19: GCRI x Internet Access



Source: The World Bank

Defense

War and conflict are among the most severe risks to global security and stability. Countries experiencing armed conflict are exposed to extreme economic, social, and humanitarian risks. Infrastructure is destroyed, social systems collapse, and populations are displaced, often leading to protracted instability

that can last for decades. War also has spillover effects, destabilizing neighboring countries, disrupting trade, and contributing to regional insecurity.

Syria, Yemen, and South Sudan are stark examples of how war can devastate a nation's stability and long-term development prospects. Conflict in these countries has led to widespread displacement, economic collapse, and humanitarian crises that have reverberated beyond their borders. Similarly, nations in the Sahel region of Africa, such as Mali and Burkina Faso, face severe security challenges from insurgent groups and cross-border terrorism. These threats have disrupted governance, displaced millions of people, and curtailed economic activity, pushing the region into a protracted crisis.

In contrast, countries that have long records of peace, such as Singapore and Switzerland, have been able to maintain high levels of economic stability and social cohesion despite global uncertainties. Countries with rocky histories that have managed to resolve conflicts, such as Colombia following its peace agreement with the FARC, have seen substantial improvements in stability, economic growth, and international standing.

Preventing conflict and promoting peace is not only a humanitarian imperative but also a critical factor in reducing global catastrophic risks. One risk mitigation measure may be to reduce military expenditures, which the IMF has categorized as "unproductive." Such expenditures are often large in relation to countries' unmet needs and yield few collateral benefits. Reductions in military spending at the national level could be reallocated to other ends, including education, public health, infrastructure, and other productivity-enhancing areas, thereby giving rise to a real "peace dividend." This raises another important issue. Our collective security mechanisms, such as those embedded in the UN Charter, are highly ineffective and have led the vast majority of countries to invest heavily in developing their respective military establishments, to protect themselves against real or often imaginary risks. Given constrained fiscal resources, this has often led to underinvestment in various dimensions of human welfare, as noted earlier in this report. A rethinking of the underpinnings of our global governance architecture would appear to be an urgent necessity in coming years, given the multiplicity of global catastrophic risks threatening our future. In this respect, the Global Governance Forum's report *A Second United Nations Charter: Modernizing the UN for a New Generation* is an important contribution to this debate.

CASE STUDY

INTERNET ACCESS IN MOROCCO & PAKISTAN

Once thought to be a privilege and luxury, internet connectivity has become a basic need for economic, political, and social participation around the world. The Internet enables access to critical services from education to healthcare, supports minority group empowerment, enhances government transparency and accountability, and develops both local and global communities. Citizens of countries with high and free levels of internet connectivity reap the benefits, while those in countries with low or metered connectivity are held back. This disparity can be seen through a comparison of two Muslim countries, Morocco and Pakistan.

With 88% of the population having access to the internet, Morocco has among the highest levels of connectivity globally, let alone among Muslim countries. Citizens have used this access to enhance social inclusion, community building, and political expression. For marginalized groups, such as the LGBTQ+ community, social media has created a virtual forum where individuals can connect, share experiences, and find support that is otherwise absent in public life due to social and legal constraints. In a country where public displays of affection, especially among same-sex couples, are socially condemned and legally punishable, online platforms have become essential in enabling the LGBTQ+ community to safely engage, organize, and advocate for their rights.

Moreover, the internet has empowered broader social and political activism in Morocco, providing a digital space for citizens to challenge authority and advocate for social justice. In 2018, the #Masatach ("I will not be silenced") campaign utilized social media to spotlight sexual harassment and called for the boycott of a prominent singer accused of rape, ultimately influencing national radio stations to stop airing his music. Similarly, in 2016, the hashtags #kulunamukhsinfikri ("We are all Mouhcine Fikri") were used to mobilize street protests after a fish vendor's death, highlighting governmental corruption and abuse. These movements underscore the internet's role in amplifying citizen voices, fostering public accountability, and advancing social reforms.

Meanwhile, Pakistan has among the lowest levels of internet access (21%). Pakistani authorities have a record of shutting down internet access in the lead-up to elections and in response to escalating tensions.

These blanket shutdowns or slowdowns disrupt democratic processes, restrict access to information for informed decision-making, and stifle economic activity.

Pakistan's economy features a high level of freelancers and small businesses that rely heavily on internet access. Service disruptions or shutdowns thus lead to substantial financial losses when a meaningful segment of the economy is unnecessarily and artificially isolated from global markets. Livia Saccardi, Interim Deputy Director for South Asia at Amnesty International, explains shutdowns "impact people's mobility, livelihood and ability to navigate through a difficult time further undermining their trust in authorities."

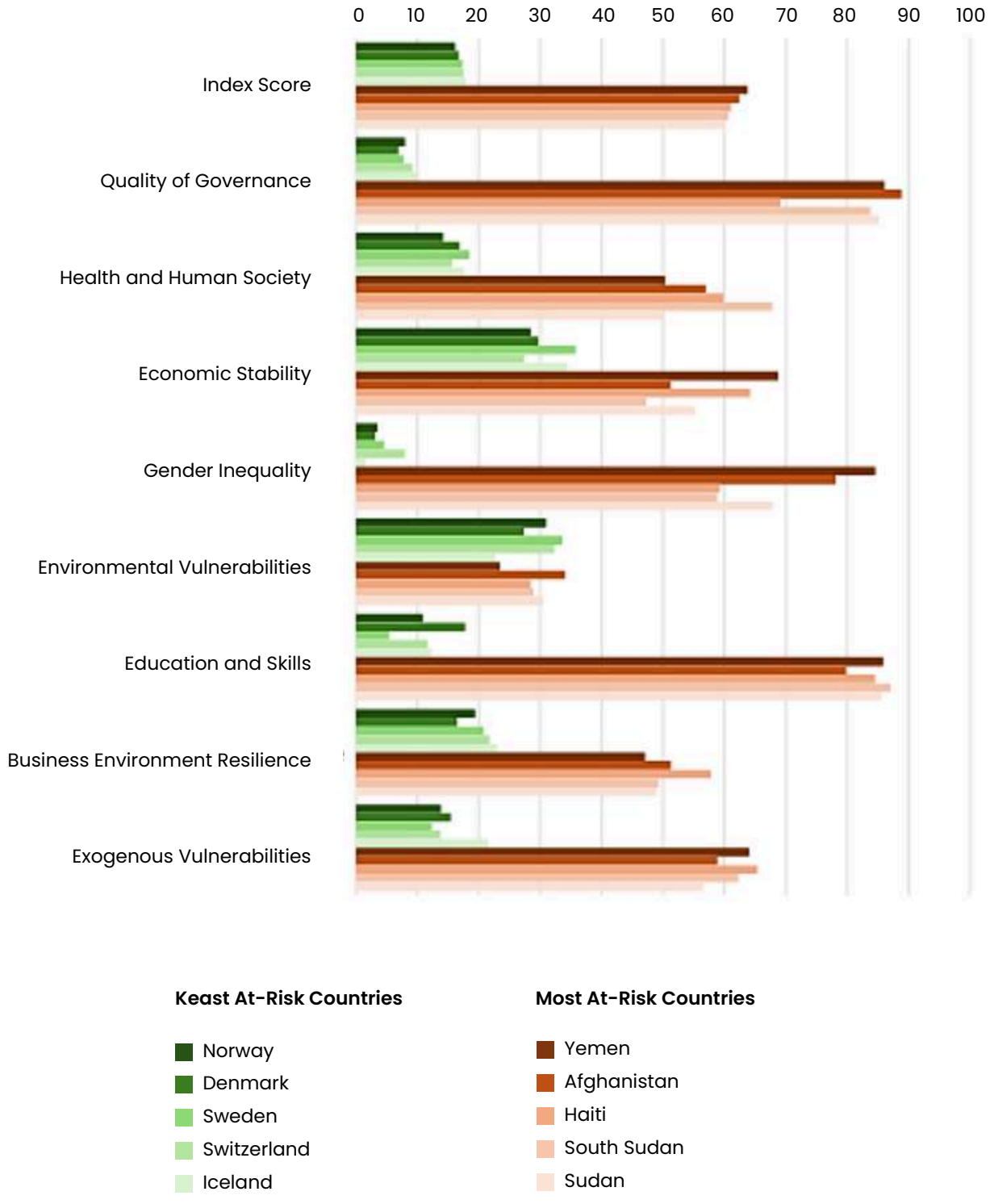
In addition to complete shutdowns, Pakistanis also face metered internet speeds. In August 2024, Pakistanis reported much slower internet over several days (estimates indicated it had fallen to less than half its usual capacity), with many claiming the slowdown coincided with the installation of a national firewall.[20] Citizens were unable to send emails or use messaging services like WhatsApp, which made it more difficult for doctors to receive statements and medical reports. Businesses were unable to contact international clients or send and receive files. Pakistan Software House Association, which represents companies specializing in information technology, estimated the slowdown would cost the country's already weak economy $300 million.

20. Hussain, Munir. "Pakistan Plans Nationwide Firewall to Monitor and Censor Internet Traffic as Elections Near." AP News, 2 Nov. 2024, apnews.com/article/pakistan-internet-disruptions-cybersecurity-firewall-elections-de408fbd514aa754fdbbf44e0f462070.

Overview of Findings

The most at-risk countries in the Global Catastrophic Risk Index are Yemen (index score of 63.8), Afghanistan (62.4), Haiti (61.1), South Sudan (60.7), and Sudan (60.0) The least at-risk countries are Norway (16.1), Denmark (16.7), Sweden (17.3), Switzerland (17.5), and Iceland (17.9). While the factors leading to these scores are complex and nuanced, the general results can be effectively illustrated by looking at the recent histories and domestic contexts of both groups.

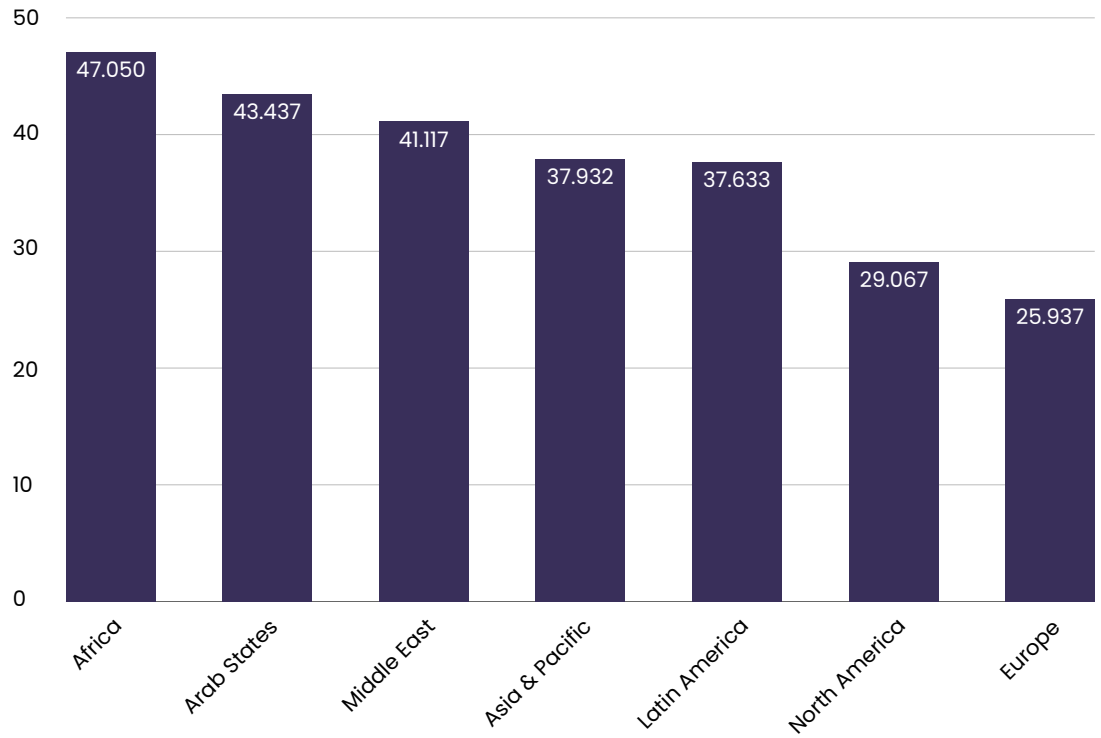
Figure 20: Most vs. Least At-Risk Countries, by pillar



The highest-risk countries have GCRI scores that are nearly four times higher than those in the lowest-risk group. Across each of these countries, political instability and outright conflict have contributed to absentee governments incapable of providing even the most basic social services. Low levels of education and basic healthcare, coupled with unpredictable business environments and significant embedded gender inequalities are the norm. In recent years, each of these low-income countries has been shaken by domestic strife: Yemen has struggled to contain domestic insurgent groups and the infiltration of regional terror cells, which led to a civil war; Afghanistan has remained war-torn in the wake of the United States' 2021 withdrawal, which sounded the death knell for the attempted establishment of a democracy; Sudan has experienced a civil war in which South Sudan became an independent state in 2011 and a military coup in 2019; and Haiti has remained the poorest country in the western hemisphere following a wave of natural disasters, gang violence, and the assassination of its president in 2021. As explained elsewhere in the report, mitigating risk and establishing resilience requires states to have the effective institutions and governance structures that to date none of these states have been able to establish.

In contrast, countries in the low-risk group are stable, high-income European states that not only have seen little violence or domestic strife in recent years but are also established democracies that put a premium on civil rights, social safety, and rule of law and where there is a high level of trust between rulers and civil society and other stakeholders. Each of these countries enjoys strong trade, cultural, and intellectual ties with their European neighbors and the global north more generally. These highly developed states see little stress of access to basic resources, have established strong social safety nets, have strong state institutions, and have low levels of corruption. In contrast to the higher-risk group of countries, citizens and governments in these countries have the luxury of focusing their attention on improving established structures rather than building them from the ground up.

Figure 21: Global Catastrophic Index Score by Region



Perhaps more surprising results can be found among those countries in the middle of the Index's score distribution. One interesting finding is that the size of the economy rankings is only weakly correlated with economic stability scores. In other words, GDP is not an end in itself as a measure of economic health. Rather, indicators including the budget deficit, national debt level, inflation rate, and labor force participation rate are all more strongly correlated with economic health. In other words, control of debt and fostering job growth through the maintenance of efficient and transparent markets will mitigate the risks facing a country more than pursuing GDP growth for its own sake.

Among the 50 largest economies, eight are among the 50 most at-risk countries according to the index (India, Egypt, Iran, Bangladesh, Pakistan, Nigeria, and Algeria, Iraq). The two largest economies, China and the United States, tell a similar story: both countries fall short in some areas while excelling in others. While China scores among the top 25 most at-risk countries by quality of governance, the United States is much safer in 138th place. Similarly, while China is the 27th most environmentally at-risk country, the United States ranks toward the middle of the pack at 77th.[21]

There is an important relationship between exposure to catastrophic risk and underground economic activity. Weak economies often push individuals to participate in informal markets for survival, and this underground economic activity further perpetuates economic instability, creating a vicious cycle. Data illustrates this stark contrast: in the 25 most at-risk countries, the underground economy is worth an average of 39.2% of GDP, compared to just 17.5% in the 25 least at-risk countries. This divergence is even more stark when examining extremes: among the ten countries with the largest underground economies relative to GDP — averaging 57.7% — the average economic stability pillar score is 56, indicating substantial vulnerability. Conversely, the ten countries with the smallest underground economies — averaging only 11.0% — show a much lower average economic stability risk score of 37, suggesting less risk, and more vibrant economic activity.

Despite the proliferation of protectionist arguments that integration with global markets exposes a country to risk, the data shows the opposite. In the 25 most at-risk countries, the average tariff rate stands at 8.2%, compared to just 1.5% in the 25 least at-risk countries. Higher levels of trade restrictiveness may be enacted to assuage deeper economic risks. Greater trade barriers however stifle economic growth by limiting access to international markets, reducing competitiveness, and artificially constraining the market for goods and services which can reduce a country's capacity to adapt to external shocks.

The data also illuminates the impact of regime type on a country's risk exposure. The average GCRI score for autocracies is 53.7, while it is 46.8 for anocracies (regimes that mix democratic with autocratic features), and 33.3 for democracies. In addition to the governmental ills characterizing many autocratic regimes — political instability, violence, corruption, and low levels of perceived government effectiveness, the data also may indicate that less democratic regimes are at greater risk across the board. This may include economic mismanagement, poor education and healthcare coverage, high levels of gender inequality, and dysfunctional business environments.

Among the 25 most at-risk countries overall, 15 are among the most at-risk countries in terms of gender inequality. Gender inequality not only diminishes women's economic participation and political representation but also weakens social cohesion, exacerbates poverty, and restricts access to education and healthcare. These structural inequities undermine the ability of societies to effectively manage risks and respond to crises, creating compounding vulnerabilities that elevate the risk of instability and reduce prospects for sustainable development.

21. The United States' ranking on environmental vulnerability reflects the balance it strikes between comparatively high-risk areas (moderate-high levels of pesticide use, a low level of renewable energy sourcing, are a large urban population) with comparatively low risk areas (low levels of air pollution, lack of widespread lowland flood risk, and a low level of internal displacement due to environmental catastrophe).

Countries that relegate women to the status of second-class citizens often exhibit deeper systemic problems that extend beyond gender inequality. In the 10 most legally restrictive countries for women, the average GCRI score is 48.1, compared to just 22.3 in the 10 least restrictive countries. This correlation holds true in countries where equitable power dynamics between men and women within households are not upheld and where violence against women is widespread. By relegating women to a subordinate status, states signal not only their stance on gender but also their broader incapacity to protect their citizens, advocate in their best interest, or defend against the multifaceted risks facing them.

Similarly, among the 25 most at-risk countries overall, 12 are among the most at-risk countries in terms of Education & Skills. Low levels of education and poorly functioning education systems are mutually reinforcing with economic, social, and political vulnerability, creating a vicious cycle that is difficult to break. Economically, a lack of education limits the development of a skilled workforce, leading to lower productivity, higher unemployment, and diminished economic growth. Socially, it perpetuates inequality, restricts social mobility, and fosters the very instability that undermines the possibility of uniformly high-quality education service delivery. Politically, low education levels weaken civic engagement, reduce the capacity for informed decision-making, and increase susceptibility to misinformation and manipulation, which can undermine the effective governance that is critical to improving education outcomes.

The indicators comprising Pillar V (Environmental Vulnerability) contain fascinating stories in their own right and illustrate the extent to which individual indicators can tell more nuanced stories than the pillar scores they comprise. For instance, among the ten most environmentally at-risk countries, two (Russia and Japan) are among the ten largest global economies, and only two (Iran and Algeria) are within the top 50 most at-risk countries across all pillars.[22] Clearly, It is not just those countries with weak economies, poor governance, and major exogenous risks that neglect to address and effectively mitigate their environmental risks. The ten most environmentally at-risk states have overall index scores ranging from 23.5 to 48.3, a span of 123 countries. This result highlights the challenge countries face in allocating resources and establishing policies to address and mitigate all of the diverse risks they face. What the ten most environmentally at-risk countries have in common is they are coastal or island nations, which makes the risk of rising sea levels, increasing global temperatures, and/or extreme weather events more acute. This risk is most pronounced for island states including Japan and the Bahamas. Four (Qatar, Bahrain, Kuwait, and Iran) are oil-producing Middle Eastern states subject to extreme heat stress, fresh water scarcity, and agricultural fragility.

The data within Pillar V also shows that it is not always those countries that are likely to be most impacted by a ravaged environment that do the most to address their environmental footprint. Comparing scores for the Environmental Vulnerabilities Pillar and the Percent of Total Energy Derived from Renewable Sources indicator, we see a strong inverse correlation. It is those countries most sensitive to climate change that use less renewable energy, while countries relatively less sensitive to the effects of climate change derive a greater percentage of their energy from renewable sources. Looking at extremes, the five most environmentally at-risk countries derive an average of just 2% of energy from renewable sources, while the five least environmentally at-risk derive an average of 81% of their energy from renewable sources.

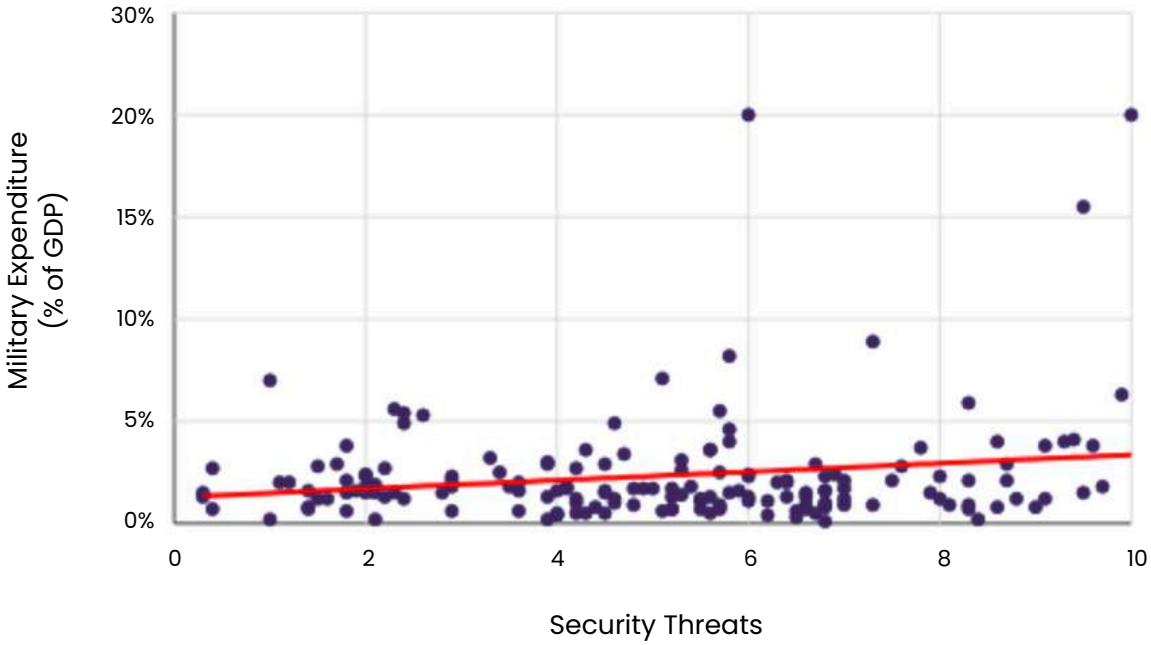
Policies misaligned with the risks they are intended to mitigate are not confined to the Environmental pillar. The same can be seen in Exogenous Risks data. Although defense risks grab headlines and global attention daily, the data show widespread failure in countries' ability to effectively mitigate them. As noted above, the IMF has called military spending "unproductive." To illustrate this statement, the data show that there is only a weak correlation between the Security Threats facing a state and the percent of

22. Japan's high environmental risk owes largely to low levels of renewable energy sourcing, a low amount of agricultural land (increasing the risk of drought, flood, or pest-related food losses), a highly urbanized population, and, being an island, high level of vulnerability to sea level change and tsunamis.

GDP a state spends on its military. This finding shows that to the extent the Security Risks indicator is an accurate proxy for the real world, countries do not allocate military resources in proportion to the risk they face, nor does an increased allocation of resources lead to diminished security threats.

Other things being equal, military spending is a contributor to budget deficits and, given the finding that risk and expenditure are only weakly related, it appears countries' military expenditures are allocating a non-negligible amount of their scarce resources to an unproductive end. Instead of investing in education, healthcare, infrastructure, or climate change mitigation technologies, countries globally are prioritizing military expenditure even though it may not actually decrease security threats.

Figure 22: Security Threats x Military Expenditure



Source: SIPRI. Fund for Peace.

Interestingly, the data shows that higher levels of post-tax subsidies[23] are associated with lower overall risk exposure. This trend suggests that subsidies can be an effective means of channeling targeted financial support to areas of greatest need, enabling countries to address crises more effectively than through market mechanisms alone. Additionally, it may indicate a certain degree of economic resilience and fiscal capacity, as only financially stable nations are able to allocate substantial resources for subsidies. Data supports this correlation: the 50 most at-risk countries allocate an average of 27.5% of their total expenditures to post-tax subsidies, whereas the 50 least at-risk countries allocate a significantly higher average of 54.0%, highlighting their ability to support key sectors and mitigate risk.

23. "Subsidies" here includes subsidies, grants, and other social benefits. These are inclusive of all unrequited, nonrepayable transfers on current account to private and public enterprises; grants to foreign governments, international organizations, and other government units; and social security, social assistance benefits, and employer social benefits in cash and in kind.

AREAS TO WATCH

1. **Conflict Spill Over:** The potential for a broader Middle East conflict is growing, driven by the Israel-Gaza war and tensions in Syria, Yemen, and Lebanon, fueled by long-standing Iranian meddling in the region. This volatility risks undermining global political stability, with repercussions for energy security, international alliances, and migration.

2. Energy Dependencies: As European countries including Germany strive to unwind their Russian gas dependency, the continent's energy security faces new challenges. Supply chain disruptions or geopolitical shifts in the Middle East could hinder efforts to secure alternative energy sources. These dynamics may leave Europe vulnerable to energy shortages, price volatility, and, as a result, economic uncertainty.

3. Will Iran go Nuclear?: A nuclear-armed Iran would change the regional – and global – balance of power in important ways. In addition to presenting direct security threats to Western-aligned countries, it could precipitate a Middle East arms race, undermining nuclear non-proliferation efforts and heightening the risk of nuclear conflict. The ripple effects could also destabilize regional alliances and increase the stakes for international security frameworks.

4. Resolution to the Russo-Ukrainian War: The trajectory of the Russo-Ukrainian war, whether toward peace or escalation, will define Europe's security landscape and influence global alliances. A peace agreement may stabilize Europe but could impact the power dynamics within NATO and other alliances. Conversely, an escalation would likely strengthen existing tensions and reinforce global divides.

5. An Agitated China: Political control at the top of the Chinese Communist Party has consolidated and the once-booming Chinese economy has shown signs of slowing down. Left with a dearth of policy options, increasingly assertive neighbors, and a possible protracted trade conflict with the United States, China's regional posture may become increasingly aggressive. Beijing's bullying of Vietnam and the Philippines, as well as its expansive territorial claims in the South China Sea could escalate and potentially culminate in an attempt to retake Taiwan — a move that would ignite a broader regional and global conflict and have deeply destabilizing effect on the global economy, including China itself given its reliance on international trade.

6. Rising Debt and Global Economic Security: Sovereign debt is on the rise globally, particularly in emerging economies. This trend, combined with growing uncertainty stemming from some unpredictability in the United States's trade policies and the risk of trade wars, could lead to a slow down in economic growth globally, destabilize trade flows and worsen the debt-carrying capacity of dozens of countries among the emerging markets and developing economies and heighten the risk of individual defaults that could destabilize entire regions.. But high-income countries are subject to pressures of their own, making the task of fiscal consolidation all the more urgent and highlighting the need for more sustainable finances, given the systemic implications of fiscal chaos in one or more large economies.

7. AI: The consequence of AI technologies' rapid rise will be more fully seen in the coming years as nations race to leverage the technology for economic, military, and strategic dominance. Disparities in development, regulation, and ethical oversight are widening, amplifying the risk of unintended consequences. AI-driven misinformation and cyber warfare threaten political stability and democracy globally, while autonomous weapon systems raise the risk of escalating conflicts.

Conclusion

The findings of this report demonstrate not only that no country is free of risk, but also that policymakers globally have largely failed to take collective action against systemic and environmental risks. Fortunately, the catastrophic risk discourse has gained traction among key actors in recent years. For example, in 2022 the US passed the Global Catastrophic Risk Management Act which aims to identify, assess, and manage extreme risks that could pose severe harm to human civilization or lead to human extinction. Under this legislation, the US government will conduct a systematic assessment of existential and global catastrophic risks, marking the world's first government-led effort to systematically evaluate the greatest threats facing humanity.

The Global Catastrophic Risk Index aligns well with this evolving focus and presents a unique picture of risk. In contrast to other reports such as the World Economic Forum's annual Global Risks Report, which is based solely on expert opinions and is therefore subjective, the GCRI relies exclusively on hard data, drawn from a myriad of sources, and accesses risk across an abundance of indicators. In addition to the overall risk rankings, which are instructive in providing a directional sense of a county's overall risk profile, the Index's individual pillar ratings can be a useful guide to where a nation's efforts and finite resources to reduce vulnerability or increase resilience can most effectively be focused.

This updated edition of the Global Catastrophic Risk Index represents an evolution and improvement on its predecessor but is by no means considered perfect. Future editions of the index will continue to broaden country coverage as far as available data allows. In addition, every indicator used in the Index will be evaluated prior to inclusion in each report in order to ensure the Index is maximally descriptive and fit for purpose. The Global Governance Forum team is invested in eliminating blind spots in order to present as comprehensive a picture of global risk as possible.

The years since the publication of the inaugural GCRI have reinforced both the need for and utility of such an index. While it is impossible to see around every corner of the future, subsequent editions of the GCRI will enable the emergence of a trajectory for countries and regions. It is our hope that this trajectory will enable governments with the early warnings and insights necessary to take action to reduce their vulnerability and increase their resilience in anticipation of oncoming challenges.

Appendix I | Data Sources

INDICATOR	SOURCE
[1.1.a] [1.1.b] [1.1.c] [1.1.d] [1.1.e]	The World Bank, Worldwide Governance Indicators, 2022
[1.1.f]	Varieties of Democracy, 2022
[1.2.a] [1.2.b] [1.3.b]	The Fund For Peace, Fragile States Index, 2024
[1.2.c]	Center for Systemic Peace, Polity5 Project, 2018
[1.2.d]	ACLED Conflict Index, 2024
[1.2.e]	The World Bank, Global State of Democracy, 2022
[1.3.a]	Transparency International, 2023
[1.3.b]	The World Bank, WEF Travel & Tourism Competitiveness Index, 2019
[1.3.c] [1.3.d] [1.4.c]	The World Bank, Global State of Democracy, 2022
[1.4.a]	Freedom House, Global Freedom Scores, 2024
[1.4.b]	Reporters Without Borders, World Press Freedom Index, 2024
[2.1.a]	WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene, (2023)
[2.1.b] [2.1.d] [2.3.e]	World Health Organization, Sustainable Development Goals, 2024
[2.1.c]	World Health Organization, Global Strategy for Women's, Children's and Adolescents' Health Monitoring Framework, 2024
[2.1.e]	World Health Organization, Universal Health Coverage Service Coverage Index, 2023
[2.1.f]	WHO/UNICEF Immunization Coverage Estimates, 2023
[2.2.a]	The World Bank, Gender Statistics, 2016
[2.2.b]	UNICEF, Joint Child Malnutrition Estimates, 2024
[2.2.d]	United Nations, Sustainable Development Goals, 2024
[2.2.f] [2.3.d]	World Health Organization, Global Health Observatory, 2024
[2.3.c]	United Nations, Inter-agency Group for Child Mortality Estimation, 2024
[2.4.a] [2.4.b] [2.4.c] [2.4.d]	United Nations Office on Drugs and Crime, 2024
[3.1.a] [3.1.b] [3.1.c]	International Monetary Fund, World Economic Outlook, 2022
[3.1.d] [3.1.e] [4.1.e]	The World Bank, International Labour Organization, 2024

Appendix I | Data Sources (cont.)

INDICATOR	SOURCE
[3.1.f]	The World Bank, International Debt Statistics, 2022
[3.2.a]	The World Bank, Center for Economic Policy Research, 2020
[3.2.b]	The World Bank, Development Research Group, 2020
[3.2.c] [3.2.f]	The World Bank, World Development Indicators, 2021
[3.2.d]	The World Bank, WEF Global Competitiveness Index, 2017
[3.2.e]	The World Bank, World Development Indicators, 2023
[3.3.a] [3.3.b] [3.3.c] [3.3.d]	International Monetary Fund, Financial Soundness Indicators, 2023
[4.1.a] [4.1.b] [4.1.c] [4.1.d] [4.2.a] [4.2.b] [4.2.c] [4.2.d] [4.3.a]	The World Bank, Women, Business and the Law, 2024
[4.2.e]	The World Bank, Inter-Parliamentary Union, 2024
[4.3.a]	The World Bank, Enterprise Surveys, 2023
[4.3.b]	The World Bank, The United Nations VAW-IAWGED, 2019
[5.1.a] [5.1.d] [5.2.a]	Food and Agriculture Organization of the United Nations, 2024
[5.1.b]	The World Bank, Global Burden of Disease, 2020
[5.1.c] [5.1.e] [5.2.c] [5.3.d] [2.2.c] [2.2.e] [2.3.a] [2.3.b] [8.1.a] [8.1.b] [8.1.c] [8.2.a] [8.2.c]	The World Bank, World Development Indicators, 2023
[5.2.b]	The World Bank, Climate Change Indicators, 2021
[5.3.a] [5.3.c]	The International Disaster Database
[5.3.b]	The World Bank, Climate Change Knowledge Portal, 2024
[5.3.e]	Disasters Global Internal Displacement Database, 2023
[6.1.a] [6.1.b] [6.1.c] [6.2.a] [6.2.b] [6.2.c] [6.2.d] [6.2.e]	The World Bank, UNESCO Institute for Statistics, 2023
[7.1.a] [7.1.b] [7.1.c] [7.1.d] [7.1.e]	The World Bank, Doing Business, 2019
[8.2.b]	United Nations, International Telecommunication Union, 2024
[8.3.a] [8.3.b]	Fund for Peace, Fragile States Index, 2024
[8.3.c]	International Campaign to Abolish Nuclear Weapons, 2024
[8.3.d]	Stockholm International Peace Research Institute, Military Expenditure Database, 2023

Appendix II | Global Catastrophic Risk Index Rankings

COUNTRY	INDEX SCORE	RANKING INDEX (most to least at-risk)
Yemen	63.8	1
Afghanistan	62.4	2
Haiti	61.1	3
South Sudan	60.7	4
Sudan	60.0	5
D.R. Congo	56.9	6
Pakistan	56.0	7
Syrian Arab Republic	55.3	8
Ethiopia	54.9	9
Mali	54.8	10
Libya	54.1	11
Bangladesh	53.7	12
Iraq	53.5	13
Niger	53.5	14
Cameroon	53.2	15
Burkina Faso	53.2	16
Lebanon	52.2	17
Myanmar	51.0	18
Mauritania	50.7	19
Nigeria	50.7	20
Guinea	50.7	21
Angola	50.0	22
Liberia	49.6	23
Burundi	49.2	24
Venezuela	49.0	25
Mozambique	48.5	26
Egypt	48.4	27

Appendix II | Global Catastrophic Risk Index Rankings (cont.)

COUNTRY	INDEX SCORE	RANKING INDEX (most to least at-risk)
Iran	48.3	28
Benin	48.0	29
Zimbabwe	48.0	30
Sierra Leone	47.5	31
Papua New Guinea	47.3	32
Madagascar	47.2	33
Tanzania	47.1	34
Uganda	46.5	35
India	45.1	36
Eswatini	44.6	37
Algeria	44.2	38
Sao Tome and Principe	44.2	39
Suriname	44.1	40
Senegal	43.6	41
Lesotho	43.5	42
Honduras	43.4	43
Belize	43.4	44
Malawi	43.1	45
Kenya	43.0	46
Ivory Coast	42.9	47
Sri Lanka	42.7	48
Solomon Islands	42.5	49
Timor-Leste	42.3	50
Maldives	42.3	51
Jordan	42.2	52
Cambodia	42.2	53
Philippines	42.2	54

Appendix II | Global Catastrophic Risk Index Rankings (cont.)

COUNTRY	INDEX SCORE	RANKING INDEX (most to least at-risk)
Lao PDR	41.9	55
Guatemala	41.8	56
Zambia	41.8	57
Nicaragua	41.4	58
Nepal	41.1	59
Indonesia	41.1	60
Bolivia	41.1	61
Ghana	40.9	62
Tajikistan	40.7	63
Morocco	40.7	64
Tunisia	40.4	65
El Salvador	40.3	66
Ukraine	40.0	67
Bahamas	39.7	68
Vanuatu	39.5	69
Fiji	39.2	70
Rwanda	38.8	71
Jamaica	38.4	72
Trinidad and Tobago	38.4	73
Grenada	38.1	74
South Africa	38.1	75
Turkey	37.9	76
Botswana	37.9	77
Uzbekistan	37.7	78
St. Lucia	37.2	79
Brazil	37.1	80
Colombia	37.1	81

Appendix II | Global Catastrophic Risk Index Rankings (cont.)

COUNTRY	INDEX SCORE	RANKING INDEX (most to least at-risk)
Paraguay	37.0	82
Namibia	37.0	83
Ecuador	36.8	84
Bahrain	36.7	85
Mexico	36.6	86
Tonga	36.4	87
Samoa	36.3	88
Cabo Verde	36.3	89
Russia	36.1	90
Guyana	36.1	91
Bosnia and Herzegovina	34.9	92
Dominican Republic	34.8	93
China	34.7	94
Kyrgyzstan	34.4	95
Panama	34.0	96
Vietnam	34.0	97
Peru	33.5	98
Saudi Arabia	33.5	99
Belarus	32.9	100
Albania	32.9	101
Thailand	32.7	102
Kuwait	32.7	103
Azerbaijan	32.7	104
Argentina	32.5	105
Barbados	32.5	106
Oman	32.2	107
Georgia	31.8	108

Appendix II | Global Catastrophic Risk Index Rankings (cont.)

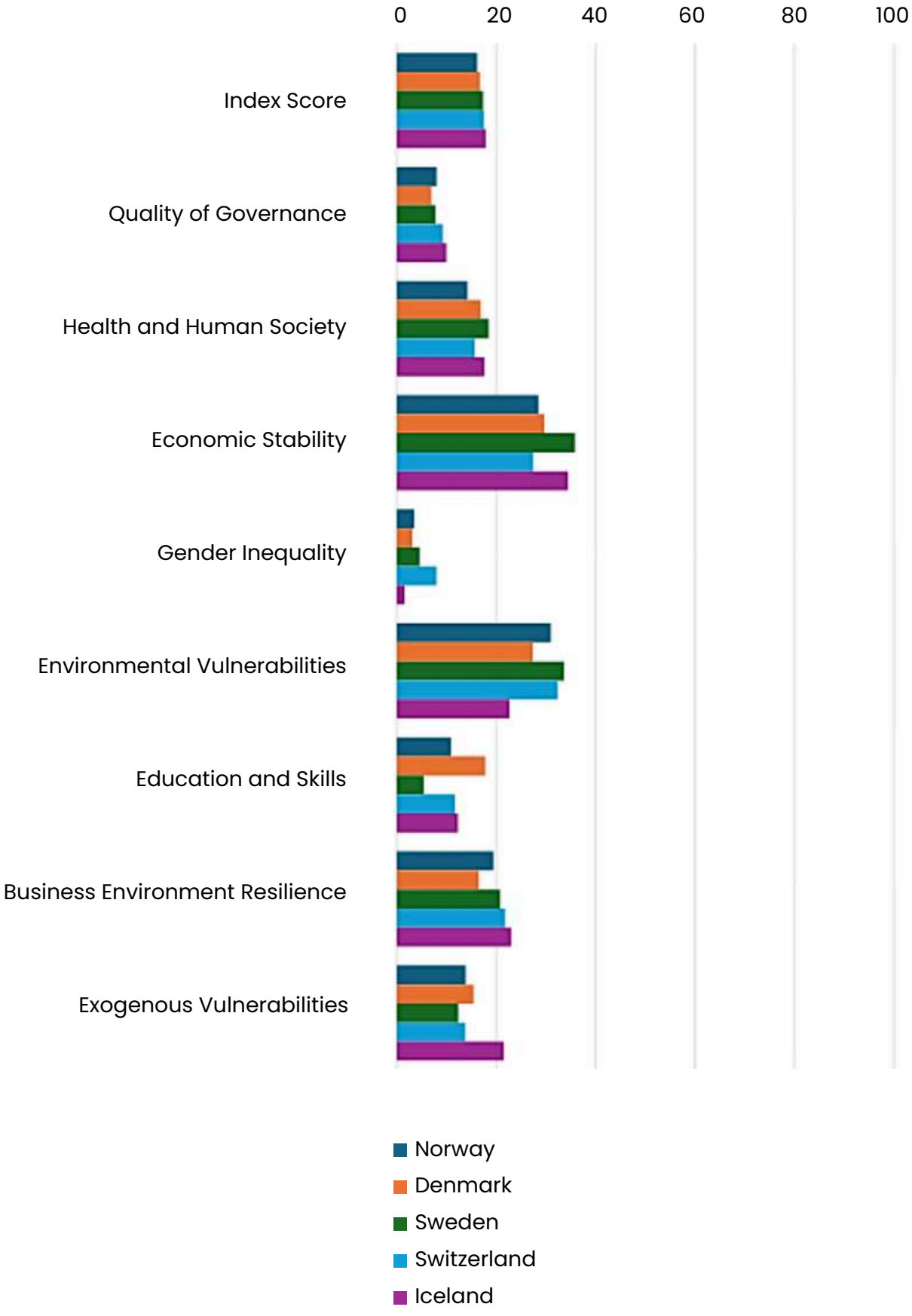
COUNTRY	INDEX SCORE	RANKING INDEX (most to least at-risk)
Greece	31.5	109
Malaysia	31.5	110
Israel	31.2	111
Serbia	31.1	112
Bhutan	31.0	113
Mauritius	30.7	114
Mongolia	30.5	115
Moldova	30.5	116
Kazakhstan	30.3	117
Brunei Darussalam	30.3	118
Armenia	30.1	119
Romania	30.1	120
Bulgaria	30.0	121
Qatar	29.9	122
Montenegro	29.7	123
North Macedonia	29.5	124
Italy	28.8	125
Poland	27.9	126
Hungary	27.9	127
Malta	27.3	128
Chile	27.2	129
Costa Rica	27.0	130
Croatia	27.0	131
Cyprus	26.4	132
France	26.2	133
United Arab Emirates	26.1	134
Japan	25.7	135

Appendix II | Global Catastrophic Risk Index Rankings (cont.)

COUNTRY	INDEX SCORE	RANKING INDEX (most to least at-risk)
United States	24.8	136
Uruguay	24.7	137
Spain	24.7	138
Belgium	24.4	139
Latvia	24.2	140
Czech Republic	24.1	141
Singapore	23.8	142
Netherlands	23.5	143
Slovakia	23.4	144
Portugal	23.3	145
Lithuania	23.0	146
Luxembourg	22.9	147
Canada	22.8	148
Korea	22.7	149
United Kingdom	22.6	150
Germany	22.5	151
Estonia	22.5	152
Slovenia	22.0	153
Austria	21.3	154
Australia	19.2	155
Finland	19.0	156
New Zealand	18.9	157
Ireland	17.9	158
Iceland	17.9	159
Switzerland	17.5	160
Sweden	17.3	161
Denmark	16.7	162
Norway	16.1	163

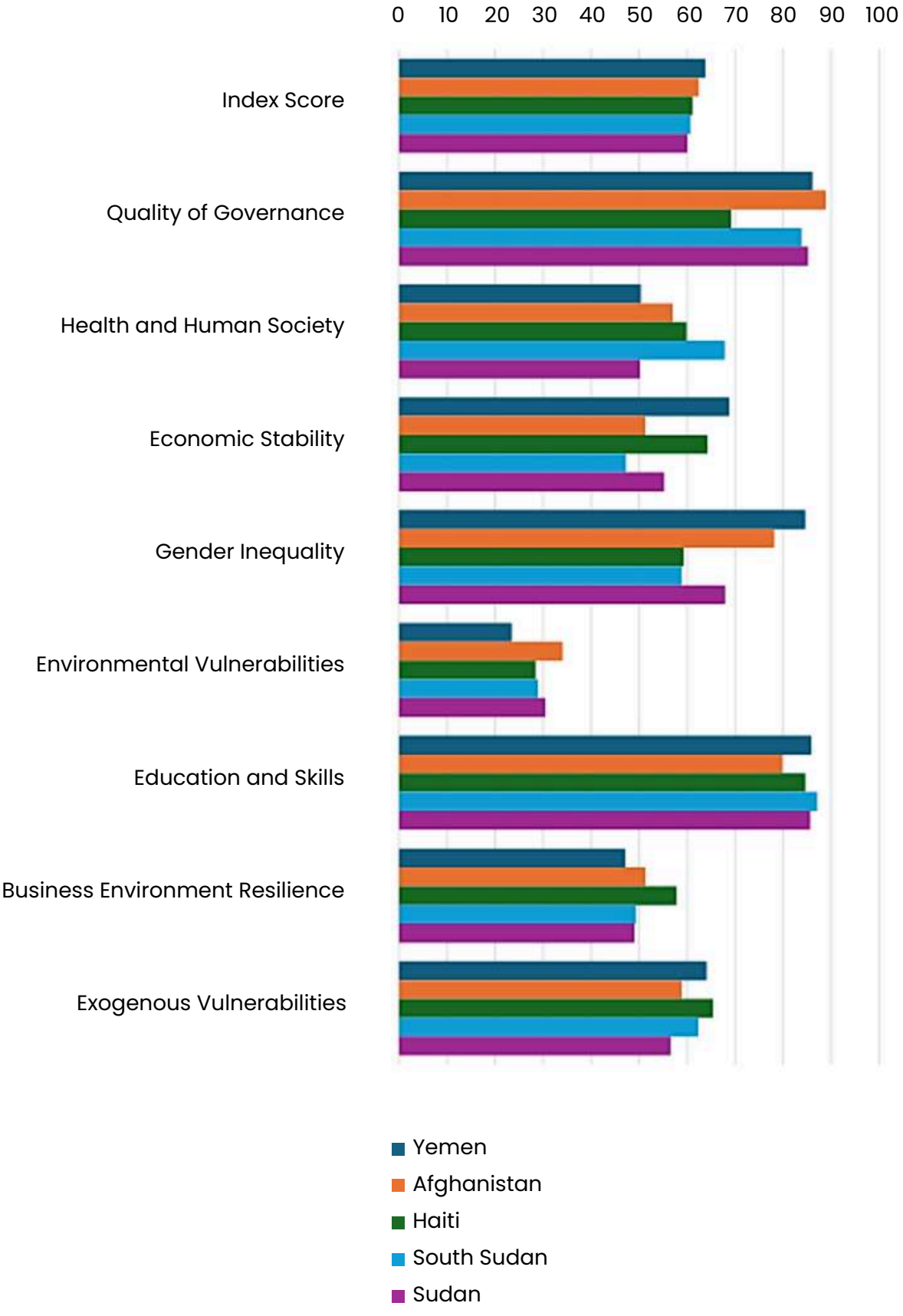
Appendix III | Pillar Scores for Most and Least At-Risk Countries

Pillar Scores for the Least At-Risk Countries



Appendix III | Pillar Scores for Most and Least At-Risk Countries (cont.)

Pillar Scores for the Most At-Risk Countries



Bibliography

ASRA. *Honest Hope: Executive Summary 2024*. ASRA, 2024. https://risk-assessment.files.svdcdn.com/production/assets/images/ASRA_Honest-Hope_ExecutiveSummary_2024.pdf?dm=1728897472.

Center on Budget and Policy Priorities. "Tracking the Recovery from the Pandemic Recession." *CBPP*, www.cbpp.org/research/economy/tracking-the-recovery-from-the-pandemic-recession.

Council on Foreign Relations. *Growing Economies Through Gender Parity. Council on Foreign Relations,* www.cfr.org/womens-participation-in-global-economy.

Council on Foreign Relations. *Venezuela: The Rise and Fall of a Petrostate.* 26 Sept. 2023, www.cfr.org/backgrounder/venezuela-crisis.

Energy Institute - Statistical Review of World Energy (2024).

Energy Policy Institute at the University of Chicago. "Most of the World Breathes Unsafe Air, Taking More Than 2 Years Off Global Life Expectancy." Air Quality Life Index (AQLI), 29 Aug. 2022, https://aqli.epic.uchicago.edu/news/most-of-the-world-breathes-unsafe-air-taking-more-than-2-years-off-global-life-expectancy/.

Eurasia Group. *Top Risks 2024: Identifying Emerging Global Challenges*. Eurasia Group, 2024. https://www.eurasiagroup.net/issues/top-risks-2024.

Feed the Future. “Niger.” Feed the Future, www.feedthefuture.gov/country/niger/

Food and Agriculture Organization of the United Nations. The Conservation of Healthy Ecosystems and Biodiversity for Livelihoods and Food Security. FAO, openknowledge.fao.org/server/api/core/bitstreams/4e57915f-a5a3-4867-a771-57cceb36230b/content.

Global Challenges Foundation. Global Catastrophic Risks 2024: Managing Risks Through Collective Action. Global Challenges Foundation, 2024. https://globalchallenges.org/app/uploads/2024/01/Global-catastrophic-risk-2024.pdf.

Global Governance Forum. The Global Catastrophic Risk Index 2022. Global Governance Forum, 2022. https://globalgovernanceforum.org/wp-content/uploads/2020/12/GGF_The-Global-Catastrophic-Risk-Index_2022.pdf.

Hussain, Munir. "Pakistan Plans Nationwide Firewall to Monitor and Censor Internet Traffic as Elections Near." *AP News*, 2 Nov. 2024, apnews.com/article/pakistan-internet-disruptions-cybersecurity-firewall-elections-de408fbd514aa754fdbbf44e0f462070.

International Monetary Fund. *Global Debt Database.* IMF, www.imf.org/en/Data/GDD.

International Organization for Migration. *World Migration Report 2024.* United Nations, 2024, publications.iom.int/books/world-migration-report-2024.

Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). Summary for Policymakers of the Thematic Assessment of the Sustainable Use of Wild Species. IPBES, files.ipbes.net/ipbes-web-prod-public-files/downloads/spm_unedited_advance_for_posting_htn.pdf.

Johns Hopkins University. "COVID-19 Dashboard by the Center for Systems Science and Engineering (CSSE)." *Coronavirus Resource Center*, coronavirus.jhu.edu/map.html

Lloyd's Register Foundation. *What the World Worries About: Global Perceptions and Experiences of Risk and Harm*. World Risk Poll, 2024. https://wrp.lrfoundation.org.uk/publications/what-the-world-worries-about-global-perceptions-and-experiences-of-risk-and-harm.

Harari, Yuval Noah. Nexus: A Brief History of Information Networks from the Stone Age to AI. Random House, 2024.

Maizland, Lindsay, and Claire Felter. "Venezuela: The Rise and Fall of a Petrostate." Council on Foreign Relations, 5 Dec. 2023, https://www.cfr.org/backgrounder/venezuela-crisis

National Oceanic and Atmospheric Administration. "U.S. Coastline to See Up to a Foot of Sea Level Rise by 2050." NOAA, 15 Feb. 2022, https://www.noaa.gov/news-release/us-coastline-to-see-up-to-foot-of-sea-level-rise-by-2050.

RAND Corporation. *Advancing the Green Economy with Circular Practices*. RAND, 2024. https://www.rand.org/pubs/research_reports/RRA2981-1.html.

RAND Corporation. Global Catastrophic Risk Assessment. RAND Corporation, 2024. https://www.rand.org/pubs/research_reports/RRA2981-1.html.

RAND Corporation. *Supporting the Transition to a Green Economy Through the Circular Economy and Sustainable Products*. RAND, 2024. https://www.rand.org/pubs/research_briefs/RBA2981-1.html.

UK Ministry of Defence. *Global Strategic Trends: Out to 2055*. Ministry of Defence, 2024. https://assets.publishing.service.gov.uk/media/673602412469c5b71dbc7b6f/Global_Strategic_Trends_Out_to_2055.pdf.

United Nations Office for Disaster Risk Reduction (UNDRR). *Building Resilience and Reducing Risks: Progress Report 2024*. UNDRR, 2024. https://www.undrr.org/media/100220/download?startDownload=20241205.

USAID. *Education and Empowerment for Girls*. United States Agency for International Development, https://2017-2020.usaid.gov/education/girls.

World Bank. "Nearly 2.4 Billion Women Globally Don't Have Same Economic Rights as Men." *World Bank*, 1 Mar. 2022, www.worldbank.org/en/news/press-release/2022/03/01/nearly-2-4-billion-women-globally-don-t-have-same-economic-rights-as-men.

World Bank. "SDG Atlas 2017: SDG 5 - Gender Equality." *World Bank*, 2017, datatopics.worldbank.org/sdgatlas/archive/2017/SDG-05-gender-equality.html.

World Economic Forum. The Global Risks Report 2024. World Economic Forum, 2024. https://www.weforum.org/publications/global-risks-report-2024/.

World Health Organization. *Preventing Disease Through Healthy Environments: A Global Assessment of the Burden of Disease from Environmental Risks.* World Health Organization, 2016.

World Health Organization. *Standards for Improving Quality of Maternal and Newborn Care in Health Facilities*. World Health Organization, 2016, apps.who.int/iris/bitstream/handle/10665/250368/9789241511131-eng.pdf?sequence=1.

About the Authors

Augusto Lopez-Claros

Augusto Lopez-Claros is an international economist and the Executive Director of the Global Governance Forum. He has published several books on global governance reform and is currently spearheading the Global Governance Forum's drafting of a Second United Nations Charter. He brings more than 30 years of experience in international organizations, including most recently as director of the Global Indicators Group at the World Bank, one of the departments within the Bank's research Vice Presidency. Previously he was chief economist at the World Economic Forum, where he directed the Global Competitiveness Program and edited the Global Competitiveness Report, the Forum's flagship publication. Before joining the Forum, he worked for several years in the financial sector in London, with a special focus on emerging markets. He was the International Monetary Fund's Resident Representative in Russia during the 1990s. He has also been a Senior Fellow at the Edmund Walsh School of Foreign Service at Georgetown University. Educated in England and the United States, he received a diploma in Mathematical Statistics from Cambridge University and a Ph.D. in Economics from Duke University. Recent publications include *"Removing Impediments to Sustainable Economic Development: The Case of Corruption"* (2015), *Equality for Women = Prosperity for All* (2018, St. Martin's Press) and Global Governance and the Emergence of Global Institutions for the 21st Century (2020, Cambridge University Press). His book Global Governance and International Cooperation: Managing Global Catastrophic Risks in the 21st Century, coedited with Richard Falk, has just been published by Routledge (2024). Augusto has lectured at some of the world's leading universities, think tanks and international organizations; a list of recent lectures can be found at: www.augustolopezclaros.com.

John Miller

John Miller is the Program Coordinator for the Global Governance Forum's Global Catastrophic Risk Initiative and has been a Fellow with the Forum since March 2021. He is a legal consultant based in Washington, DC, where he counsels Am Law 100, Global 100, and smaller specialty and regional firms on strategy. John graduated *cum laude* from Georgetown University's School of Foreign Service in 2019 with a BS in International Political Economy, for which his thesis examined the interaction between the formal and informal economic sectors in Latin America. He has also completed a certificate program in Diplomacy Studies from the Institute for the Study of Diplomacy, with a research focus on the constitutional impact of balancing sovereign policy space with delegation to international institutions. From 2018 to 2019, he served as an Editor-In-Chief of the Georgetown Journal of International Affairs, the School of Foreign Service's flagship publication.

Other Contributors

Aminah Mallik

Aminah Malik is pursuing a degree in International Political Economy at Georgetown University. Her professional interests include economic research and policy research and advising.

Benjamin Sweeney

Benjamin Sweeney is a student at Georgetown University's College of Arts & Sciences majoring in Economics and History. Ben is passionate about working with non-profit organizations and community leaders to address economic and political issues.


Global Governance Forum
Global Governance Forum